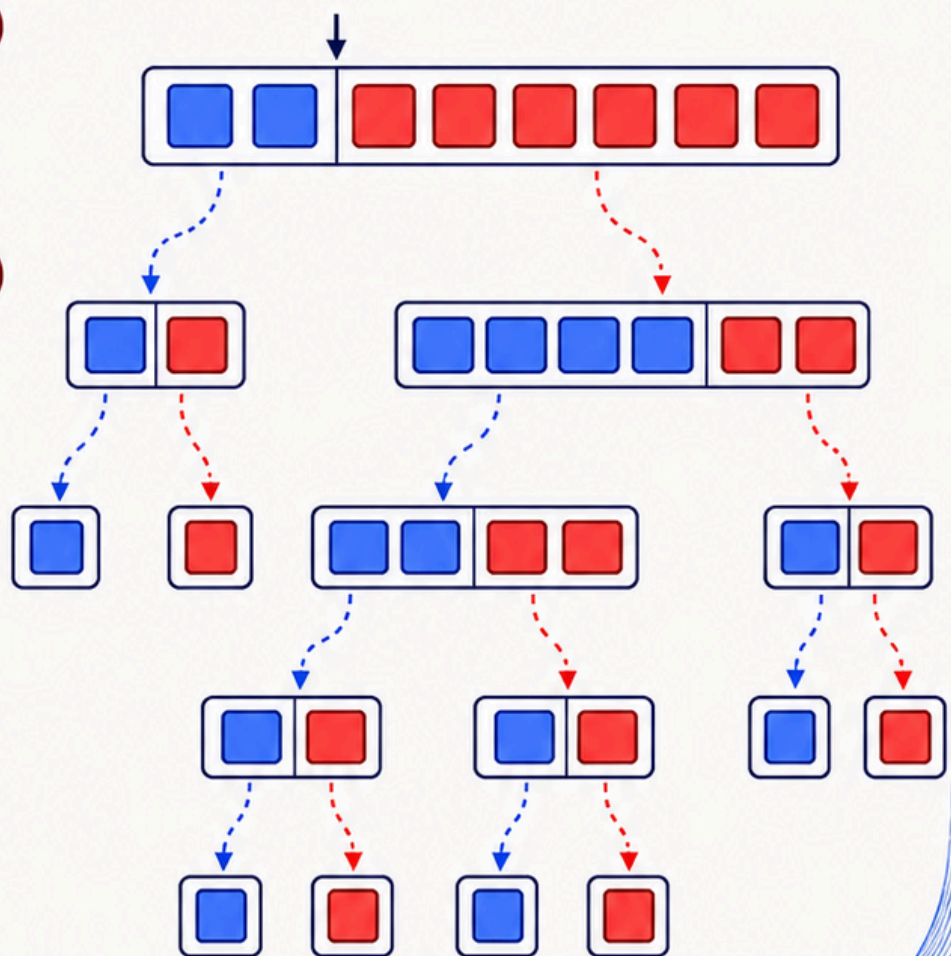
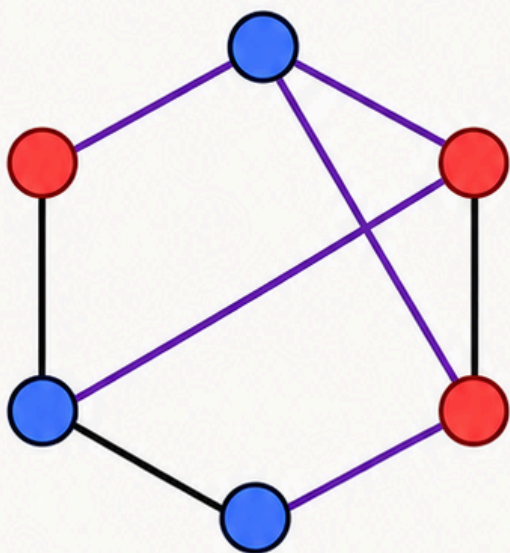


Randomised Algorithms

Sahil M Deo



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Chapter 1

Introduction

1.1 What even is a Randomised Algorithm?

A randomised algorithm is an algorithm that makes random choices during its execution. Unlike deterministic algorithms, whose behaviour is completely determined by the input, the behaviour of a randomised algorithm may vary between different executions on the same input.

Typically, the algorithm uses a random number generator (**rng**) to decide between multiple possible actions. As a result, quantities such as running time¹, memory usage, or even correctness² can become random variables (though usually not all of them at the same time).

The performance of a randomised algorithm is therefore analysed using probability and expectation. If T_{run} denotes the runtime of the algorithm, we are usually interested in quantities such as $T_{algo} = E[T_{run}]$ over all possible random choices made by the algorithm.³

Randomised algorithms are often used for the following purposes:

- avoid adversarial worst case inputs (common in competitive programming)
- simplify algorithm design and implementation
- obtain polynomial-time approximate solutions to NP problems (the field of heuristic algorithms)

1.2 Implementation of Randomness in C++

Let us make things more concrete with some actual code and implementation details! **C++** provides pseudo-random⁴ number generators through the `<random>` library. A commonly used generator is `mt19937`, which is based on the Mersenne Twister algorithm.

¹Such algorithms are called Las Vegas algorithms.

²Such an algorithm is called a Monte Carlo algorithm. The fact that correctness itself may become probabilistic can initially seem undesirable, especially in critical systems where absolute correctness is expected. However, in many practical situations, a sufficiently small error probability is considered acceptable. For example, if an algorithm can be shown to fail with probability at most 10^{-30} , then the chance of failure may already be smaller than the probability of random hardware faults such as memory bit flips caused by cosmic radiation, which programmers typically ignore during ordinary software development.

³ T_{algo} can still be a random variable over the input - so this may **not** be the same as average case performance of the algorithm T_{avg} , which is an expected value over all possible inputs. But for cleanliness of presentation, we will often ignore this distinction and use these terms interchangeably, more often taken to mean T_{avg} than T_{algo} .

⁴Since computers are deterministic, they rely on parameters like atmospheric noise and temperature for randomness. The details are irrelevant for us... We will take pseudo random as true random for our purposes

```
#include <bits/stdc++.h>
mt19937 rng(chrono::steady_clock::now().time_since_epoch().count()); //using time as a seed
```

This creates an object `rng` with seed⁵ as current time which will be passed as a parameter to other functions. It has the capability to produce "truly" random 32-bit integers.

Generating Random Integers

To generate a random integer uniformly from a range $[L, R]$, we use

```
uniform_int_distribution<int>uid(L,R);

int x=uid(rng);
int y=uid(rng);
.
.
.
```

Each generation takes around constant time.

Generating Random Real Numbers

To generate a real number uniformly from an interval $[x, y)$, we use

```
uniform_real_distribution<double>urd(x,y);

double x=urd(rng);
double y=urd(rng);
.
.
.
```

This also takes constant time per generation.

Generating a Random Permutation

A random permutation of an array (in-place modification) can be generated using `shuffle`:

```
shuffle(a.begin(),a.end(),rng);
```

Internally, this does something like:

```
for(int i=n-1;i>0;i--)
{
    int j=random integer from [0,i];
    swap(a[i],a[j]);
}
```

Deduce why all permutations are equally likely with this method!

⁵Seed values are used in pseudo random generators as a way to recreate the generated random numbers. This can be used for debugging a randomised algorithm. If you don't care about that, set it to the current time in milliseconds or something similar and that should be good enough!

Implementing Random Decisions

Sometimes, an algorithm must choose between multiple actions with specified probabilities. For example, let three actions X,Y,Z be such that they are executed with probabilities:

$$\Pr(X) = \frac{1}{5}, \quad \Pr(Y) = \frac{3}{5}, \quad \Pr(Z) = \frac{1}{5}$$

One simple way to implement this is to generate a random integer uniformly from 1 to 5:

```
uniform_int_distribution<int>uid(1,5);
```

```
int v=uid(rng);
```

```
if(v==1)
{
    //do X
}
else if(v<=4)
{
    //do Y
}
else
{
    //do Z
}
```

thus implementing the required probabilities.

Chapter 2

Algorithmic Performance

For algorithms whose performance¹ depends on the input size or other input parameters, we usually estimate the number of elementary operations performed as a function of the input size n or the maximum value of input A_{max} , usually denoted by A . This gives a mathematical way to compare algorithms independent of hardware or implementation details.

A classical example is Bubble Sort. The algorithm repeatedly traverses the array, comparing adjacent elements and swapping them whenever they are in the wrong order. After each pass, the largest remaining element moves to its correct position near the end of the array.

With some thought it can be proved that the total number of swaps performed by Bubble Sort is exactly equal to the number of inversions N_{inv} in the input array. (An inversion is a pair of indices (i, j) such that $i < j$ but $a_i > a_j$.) By taking swaps to be one operation taking constant time, we can estimate runtime of Bubble sort to be $T_{run} \propto N_{inv}$

2.1 Worst Case Analysis

In worst case analysis, we determine the maximum possible number of operations over all valid inputs of size n . For Bubble Sort, the worst case occurs when the array is sorted in reverse order, where every pair is inverted.

$$N_{inv,max} = \binom{n}{2} = \frac{n(n-1)}{2}$$

Worst case analysis is useful because it guarantees an upper bound on the running time regardless of the input provided.

2.2 Average Case Analysis

In average case analysis, we compute the expected number of operations assuming that all valid inputs are equally likely.

For Bubble Sort, this corresponds to analysing the expected number of inversions in a random permutation of size n .

Define the indicator² variable I_{ij} for each pair (i, j) as follows:

¹Although this section focuses primarily on running time, the same methods of analysis can also be applied to other resources such as memory usage.

²Refer Appendix A.5

$$I_{ij} = \begin{cases} 1, & \text{if } (i, j) \text{ is an inversion} \\ 0, & \text{otherwise} \end{cases}$$

For every pair of indices (i, j) with $i < j$, the probability that the pair forms an inversion is

$$\Pr((i, j) \text{ is an inversion}) = \frac{1}{2} \implies \mathbb{E}[I_{ij}] = \frac{1}{2}$$

The point of defining the indicator variable is to notice:

$$N_{\text{inv}} = \sum_{1 \leq i < j \leq n} I_{ij}$$

Using linearity of expectation³,

$$\begin{aligned} \mathbb{E}[N_{\text{inv}}] &= \mathbb{E} \left[\sum_{1 \leq i < j \leq n} I_{ij} \right] = \sum_{1 \leq i < j \leq n} \mathbb{E}[I_{ij}] \\ \implies \mathbb{E}[N_{\text{inv}}] &= \sum_{1 \leq i < j \leq n} \frac{1}{2} = \frac{1}{2} \binom{n}{2} = \frac{n(n-1)}{4} \quad \left(\text{Using } \mathbb{E}[I_{ij}] = \frac{1}{2} \right) \end{aligned}$$

So notice that if we randomise the input sequence, T_{algo} becomes equal to T_{avg} thus improving performance by a factor⁴ of 2.

2.3 Time Complexity Analysis

We are sometimes interested in comparing runtimes like $T_{\text{run}}(n) = n^2$ to $T_{\text{run}}(n) = 100n \log n$. To compare such functions of n , we use asymptotic notation:

$$T_{\text{run}}(n) = \Theta(f(n))$$

which means that T_{run} grows asymptotically at the same rate as $f(n)$.

My favourite way to think about time complexity is as follows: we try to find the “simplest” function $f(n)$ such that

$$\lim_{n \rightarrow \infty} \frac{T_{\text{run}}(n)}{f(n)} = k, \quad k \in \mathbb{R}_{>0}$$

That is, $T_{\text{run}}(n)$ and $f(n)$ grow at the same asymptotic rate up to a positive constant factor. For example, Bubble Sort performs on the order of n^2 operations in both the average and worst case, so its time complexity is

$$T_{\text{run}}(n) = \Theta(n^2)$$

Asymptotic analysis allows us to compare algorithms by their limiting behaviour rather than implementation-dependent constants.

³Refer Appendix A.1

⁴Randomising the input itself takes about n operations, so more rigorously we would say “an asymptotic factor of 2” rather than an exact factor of 2. In most cases we will be able to neglect the time required to randomise the input.

This allows us to say that A: $T_{run} = n^2$ is worse than B: $T_{run} = 100n \log n$ since A is $\Theta(n^2)$ and B is $\Theta(n \log n)$ even though for some small n the first one may well be better, but for small n the performance doesn't really⁵ matter anyway...

⁵Integer multiplication is an interesting case. The classic method of multiplying digit by digit as taught in school is $\Theta(n^2)$ time, but there are more asymptotically efficient methods which can do it in $\Theta(n \log n)$ time. But for reasonably large integers also the constant factor is large enough that the school method is faster!

Chapter 3

The Classics

3.1 QuickSort

In general, for an input array of size n – to sort deterministically using only pairwise comparisons, it can be proved via a decision tree that we need at least $n \log_2 n$ operations. From a time complexity point of view this is $\Theta(n \log n)$ time. `QuickSort` does not really improve upon that, but in practice it is very quick with a small constant factor. The STL library of `C++` uses `QuickSort` in its `std::sort()` function.

3.1.1 A Simple Algorithm

Consider the following pseudocode for `QuickSort`:

```
QuickSort(A)
{
    choose an element p from A as the "pivot"
    partition A into A<p, A=p, A>p
    return {QuickSort(A<p) : (A=p) : QuickSort(A>p)}
}
```

Note that $A < p$ is our notation for the elements of A that are strictly less than p . Similarly for $A > p$ and $A = p$.

A deterministic implementation may always choose the first or last element as pivot. Unfortunately, this can lead to very poor performance on adversarial inputs. For example, if the array is already sorted and we always choose the leftmost element as pivot, the recursion becomes extremely unbalanced:

$$(n-1) + (n-2) + \dots + 1 = \Theta(n^2)$$

Thus `QuickSort` has worst case complexity

$$T(n) = \Theta(n^2)$$

The standard randomised version instead chooses the pivot uniformly at random.

```
uniform_int_distribution<int>uid(0,a.size()-1);
int pivot=uid(rng);
swap(a[pivot],a[r]);
```

This tiny modification dramatically changes the behaviour of the algorithm. Intuitively, it becomes very unlikely that we repeatedly choose extremely bad pivots.

3.1.2 Expected Runtime

Recurrence Relation

Let $T(n)$ denote the expected runtime of randomised **QuickSort** on an array of size n over all possible inputs.

Suppose the pivot finally ends up at position k , where $0 \leq k \leq n-1$. Then the recursive subproblems have sizes k and $n-1-k$. Since partitioning itself takes n operations, we obtain the recurrence

$$T(n) = T(k) + T(n-1-k) + n$$

Because the pivot is chosen uniformly at random, every value of k is equally likely. Taking expectation over all random choices made by the algorithm gives

$$T(n) = n + \frac{1}{n} \sum_{k=0}^{n-1} (T(k) + T(n-1-k))$$

$$T(n) = \Theta(n) + \frac{2}{n} \sum_{k=0}^{n-1} T(k) \quad \left(\text{Using } \sum_{k=0}^{n-1} T(k) = \sum_{k=0}^{n-1} T(n-1-k) \right)$$

Changing the variable from $T(n)$ to $S(n)$ gives,

$$\implies S(n) - S(n-1) = cn + \frac{2}{n} S(n-1) \quad \left(\text{Substituting } S(n) = \sum_{k=0}^n T(k) \right)$$

$$\implies S(n) = \left(1 + \frac{2}{n} \right) S(n-1) + cn$$

$$\implies \frac{S(n)}{(n+1)(n+2)} = \frac{S(n-1)}{n(n+1)} + \frac{cn}{(n+1)(n+2)}$$

$$\implies \frac{S(n)}{(n+1)(n+2)} - \frac{S(n-1)}{n(n+1)} = \frac{cn}{(n+1)(n+2)}$$

Summing this recurrence telescopically gives

$$\frac{S(n)}{(n+1)(n+2)} = \Theta(\log n)$$

$$\implies S(n) = \Theta(n^2 \log n)$$

$$\implies T(n) = S(n) - S(n-1) = \Theta(n \log n)$$

Hence the expected runtime of randomised **QuickSort** is $T(n) = \Theta(n \log n)$

Indicator Variables

Observe that any pair of elements is compared at most once during the entire execution of **QuickSort**.

Consider the indicator variable,

$$I_{ij} = \begin{cases} 1, & \text{if elements } i, j \text{ are compared} \\ 0, & \text{otherwise} \end{cases}$$

Then the total number of comparisons C is given by

$$C = \sum_{1 \leq i < j \leq n} I_{ij}$$

Two elements are compared only if one of them becomes the first pivot chosen among all elements lying between them in sorted order. Since all of them are equally likely, we have a 2 out of $j - i + 1$ chance of this happening.

$$\Pr(I_{ij} = 1) = \frac{2}{j - i + 1} \implies \mathbb{E}[C] = \sum_{1 \leq i < j \leq n} \frac{2}{j - i + 1} = \Theta(n \log n)$$

3.2 QuickSelect

The Problem: Find the k -th smallest element¹ of an unsorted input array.

An obvious method is to sort the array using Merge Sort² or **QuickSort** and then access the k -th element in $\Theta(1)$. But since we only care about one element, it seems inefficient to sort the entire array.

The Solution:

We pick a pivot p and partition the array as before. Basis the size of the partitions we can figure out which partition the k -th element is in. If it is in the $A=p$ partition, then we already have our answer! Else we recurse into that partition and continue this process until we find the k -th element.

```
QuickSelect(A,k)
{
    choose an element p from A as the "pivot"
    partition A into A<p, A=p, A>p
    if size(A<p) >= k
        return QuickSelect(A<p,k)
    else
    {
        if size(A<p)+size(A=p) >= k
            return p
        else
            return QuickSelect(A>p,k-size(A<p)-size(A=p))
    }
}
```

¹This is commonly referred to as the k -th order statistic of the array.

²Merge Sort is a divide and conquer deterministic sort algorithm which runs in $\Theta(n \log n)$ time.

Intuitively, this is similar to **QuickSort**. However, unlike **QuickSort**, we recurse into only one side – each step discarding a significant fraction³ of the array. This is a similar approach to Binary Search although not quite same since this does not assume a sorted array.

3.2.1 Expected Runtime

Let X_i denote the size of the search space after i recursive calls of **QuickSelect**. ($X_0 = n$)

At each step, a pivot is chosen uniformly at random and the array is partitioned around it. Depending on the rank we are searching for, **QuickSelect** recurses into exactly one of the two partitions. To obtain a worst-case bound, we imagine that the algorithm always recurses into the larger partition.

If the current search space has size x , and the pivot ends up at position k , then the larger partition has size $\max(k-1, x-k)$.

Averaging over all pivot positions,

$$\begin{aligned}\mathbb{E}[X_{i+1} \mid X_i = x] &= \frac{1}{x} \sum_{k=1}^x \max(k-1, x-k) \leq \frac{3}{4}(x-1) \leq \frac{3}{4}x \\ \implies \mathbb{E}[X_{i+1} \mid X_i] &\leq \frac{3}{4}X_i\end{aligned}$$

Taking expectations and applying the law of total expectation,

$$\begin{aligned}\mathbb{E}[X_{i+1}] &= \mathbb{E}[\mathbb{E}[X_{i+1} \mid X_i]] \leq \frac{3}{4}\mathbb{E}[X_i] \\ \implies \frac{\mathbb{E}[X_{i+1}]}{\mathbb{E}[X_i]} &\leq \frac{3}{4} \implies \frac{\mathbb{E}[X_i]}{\mathbb{E}[X_0]} \leq \left(\frac{3}{4}\right)^i \\ \implies \mathbb{E}[X_i] &\leq n\left(\frac{3}{4}\right)^i \quad (\text{Using } \mathbb{E}[X_0] = X_0 = n)\end{aligned}$$

Let T be the random variable which denotes the number of recursive calls of **QuickSelect**. It is the first i such that $X_i \leq 1$.

Since the total runtime is proportional to the number of operations (taking constant of proportionality as 1 for simplicity) which are comparisons with the pivot in this case, we have

$$\begin{aligned}E[T_{run}] &\leq \sum_{i=1}^T n\left(\frac{3}{4}\right)^i \leq \sum_{i=1}^{\infty} n\left(\frac{3}{4}\right)^i \\ \implies E[T_{run}] &\leq n \cdot \frac{\frac{3}{4}}{1 - \frac{3}{4}} = 3n\end{aligned}$$

So the expected runtime is $\Theta(n)$.

³On an expected basis, of course

3.2.2 Expected Recursion Depth

Recall that T is the first i such that $X_i \leq 1$. Since $X_i \leq 1$ is equivalent to $i \geq T$, $i < T$ is equivalent to $X_i \geq 2$.

$$\implies \Pr(i < T) = \Pr(X_i \geq 2) \leq \frac{\mathbb{E}[X_i]}{2} \quad (\text{Using Markov's inequality})$$

Upper bounding $\mathbb{E}[X_i]$,

$$\Pr(X_i \geq 2) \leq \frac{\mathbb{E}[X_i]}{2} \leq \frac{n}{2} \left(\frac{3}{4}\right)^i$$

Using the tail-expectation identity,

$$\begin{aligned} \mathbb{E}[T] &= \sum_{i=0}^{\infty} \Pr(T > i) \\ \implies \mathbb{E}[T] &\leq \sum_{i=0}^{\infty} \min\left(1, \frac{n}{2} \left(\frac{3}{4}\right)^i\right) \quad (\text{Using the trivial } \Pr(T > i) \leq 1) \end{aligned}$$

The Splitting⁴ Technique We split the sum into two parts, and bound them or find them separately, thus finding our overall bound. Here we split according to when the $\min()$ changes from 1 to $\frac{n}{2} \left(\frac{3}{4}\right)^i$.

$$\begin{aligned} \mathbb{E}[T] &\leq \sum_{i=0}^{m-1} 1 + \sum_{i=m}^{\infty} \frac{n}{2} \left(\frac{3}{4}\right)^i = m + \sum_{i=m}^{\infty} \frac{n}{2} \left(\frac{3}{4}\right)^i \\ &\quad \left(\text{where } m = \left\lceil \log_{\frac{4}{3}} \left(\frac{n}{2}\right) \right\rceil\right) \end{aligned}$$

For the summation we can factor out the first term,

$$\sum_{i=m}^{\infty} \frac{n}{2} \left(\frac{3}{4}\right)^i = \frac{n}{2} \left(\frac{3}{4}\right)^m \sum_{j=m}^{\infty} \left(\frac{3}{4}\right)^{j-m}$$

$$\text{We have } \frac{n}{2} \left(\frac{3}{4}\right)^m \leq 1 \quad (\text{By definition of } m) \quad \text{and} \quad \sum_{j=m}^{\infty} \left(\frac{3}{4}\right)^{j-m} = \sum_{j=0}^{\infty} \left(\frac{3}{4}\right)^j = \frac{1}{1 - \frac{3}{4}} = 4$$

$$\implies \sum_{i=m}^{\infty} \frac{n}{2} \left(\frac{3}{4}\right)^i \leq 4$$

Combining the two bounds,

$$\mathbb{E}[T] \leq m + 4 = \left\lceil \log_{4/3} \left(\frac{n}{2}\right) \right\rceil + 4$$

So we can conclude⁵ $\boxed{\mathbb{E}[T] = \Theta(\log n)}$

3.3 Problems

- 1 (a) We concluded an upper bound for the recursion depth of `QuickSelect`. But to justify the complexity as $\Theta(\log n)$, show that the recursion depth is lower bounded by $\Theta(\log n)$.

⁴I feel this is useful and common enough to warrant a name!

⁵Or can we? See Problems!

Chapter 4

Hashing

4.1 What even is Hashing?

A recurring theme in randomized algorithms is the idea of replacing a large object by a much smaller *fingerprint* or *hash* which approximately preserves equality.

Instead of comparing two complicated objects directly, we compare their hashes. If the hashes differ, the objects are certainly different. If the hashes are equal, the objects are *probably* equal.

The power of hashing comes from the fact that computing and comparing hashes is often significantly faster than comparing the original objects themselves. It may also be seen as a form of compression that preserves some information about the original object. In data structures like the **unordered set**, hashing helps to achieve constant time complexity (on average) within finite memory constraints.

Usually hashes are many-one functions, meaning that multiple different objects can have the same hash. Two inputs having the same hash is called a *collision*. The probability of collision depends on the specific hash function used and the distribution of inputs. A good hash function should minimize the probability of collisions for typical inputs.

4.2 Polynomial Multiplication

Suppose we are given two expressions:

$$P(x) = \prod_{i=1}^k (a_i x + b_i), Q(x) = \sum_{j=0}^k c_j x^j$$

We would like to verify whether

$$P(x) \equiv Q(x)$$

Naively expanding the product takes $\Theta(k^2)$ time, and then comparing the resulting polynomials takes $\Theta(k)$ time, so the total verification time is $\Theta(k^2)$.

A more efficient approach is to compute the hashes of the polynomials. Choose three random values

$$r_1, r_2, r_3$$

from the domain of the polynomial.

Define the hash of a polynomial to be the tuple

$$(P(r_1), P(r_2), P(r_3))$$

Similarly compute

$$(Q(r_1), Q(r_2), Q(r_3))$$

If the hashes differ at any coordinate, then the polynomials are certainly different.
 If all three evaluations agree, then with high probability the polynomials are identical.

The reason this works is that the nonzero polynomial $P(x) - Q(x)$ of degree at most k can have at most k roots in the domain from which we are choosing the random values.

If we are choosing random 32-bit integers, then the probability of a false positive (i.e. the polynomials are different but the hashes match) is at most $\left(\frac{k}{2^{32}}\right)^3$ since we are checking three independent random values, which is astronomically small for any reasonable k !

Each evaluation of P or Q at a random point takes $\Theta(k)$ time, so the total verification time is $\Theta(k)$. So we have achieved a significant improvement over the naive¹ $\Theta(k^2)$ time.

Note that choosing 3 random values has no particular significance - we could choose any small constant number of random values to achieve this - it is a tradeoff between time and correctness probability.

4.3 Matrix Multiplication

Suppose we are given matrices $A, B, C \in \mathbb{R}^{n \times n}$ and wish to verify whether

$$AB = C$$

Computing the product explicitly requires matrix multiplication. Standard matrix multiplication takes $\Theta(n^3)$ time.

time, while the fastest known algorithms run in roughly $\Theta(n^{2.37})$ time, though they are highly complicated and rarely used in practice.

4.3.1 Freivalds' Algorithm

Choose a random vector $X \in \{0, 1, \dots, p-1\}^n$, aka of form

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Instead of comparing the matrices themselves, we compute ABX and CX . Since matrix-vector multiplication takes only $\Theta(n^2)$ and we are doing 3 such multiplications – $B \cdot X$, then $A \cdot (BX)$, and $C \cdot X$. Then we compare the two products entry by entry in $\Theta(n)$ time. So the overall procedure is $\Theta(n^2)$ time.

The quantity MX may be viewed as a compact randomized hash of the matrix M . If two matrices differ, their hashes are unlikely to coincide for a random choice of X .

More generally, one may choose k independent random vectors

$$X_1, X_2, \dots, X_k$$

and define the hash of a matrix M as

$$\mathbf{H}(M) = (MX_1, MX_2, \dots, MX_k)$$

Each additional random vector independently decreases the probability of error. Here k must be $\Theta(1)$ to preserve the overall $\Theta(n^2)$ time complexity.

¹There exists deterministic techniques to get this down to $\Theta(k \log k)$ time, but they have a large constant factor and also are very difficult to implement.

Error Probability Analysis

Let $M = AB - C$. We want to bound the probability that $MX = 0$ when M is nonzero. Choose a prime p , and consider the finite field

$$\mathbb{Z}_p = (\{0, 1, 2, \dots, p-1\}, +_p, \times_p)$$

To avoid making the entries of $M \equiv 0 \pmod{p}$ when it is nonzero over $\mathbb{Z}$ (and thus avoiding making it null in $\mathbb{Z}_p$), we use such a p which is strictly greater than every entry of M . We then choose the random vector $X \in \mathbb{Z}_p^n$ uniformly at random.

Notice that whenever $MX = 0$ over the integers, we also have $MX = 0$ over $\mathbb{Z}_p$ since $0 \equiv 0 \pmod{p}$. So the set of false positives over $\mathbb{Z}$ is a subset of the set of false positives over $\mathbb{Z}_p$. Thus the probability of a false positive over $\mathbb{Z}$ is at most the probability of a false positive over $\mathbb{Z}_p$.

Now $(\mathbb{Z}_p)^n$ is a vector space² of dimension n over $\mathbb{Z}_p$. If M is not the null matrix, the dimension of its nullspace³ is at most $n - 1$.

A d -dimensional vector space over $\mathbb{Z}_p$ contains exactly p^d vectors. So

$$|\text{nullspace}(M)| \leq p^{n-1}$$

Since X is chosen uniformly from $(\mathbb{Z}_p)^n$, which contains p^n vectors,

$$\Pr(MX = 0 \mid \mathbb{Z}) \leq \Pr(MX = 0 \mid \mathbb{Z}_p) \leq \frac{p^{n-1}}{p^n} = \frac{1}{p}$$

A suitable choice is $p = 2^{64} + 13$ if we are working with 64-bit integers, which gives an astronomically small error probability of about 5.4×10^{-20} .

4.4 A Matrix Problem

Problem Statement⁴: Given two $n \times m$ matrices A and B , determine if they are equivalent under row and column permutations. That is, can we permute the rows and columns of A to obtain B ?

Additional Constraints: A and B both satisfy – the set of all entries is $\{1, 2, \dots, nm\}$.

Reduction: First we reduce the problem to a simpler one. If we consider $R(M)$ to be the set of all R_i where each R_i is the set of values in the i -th row of a matrix M , and similarly $C(M)$ then the matrices A and B are equivalent under row and column permutations if and only if $R(A) = R(B)$ and $C(A) = C(B)$.

Proof of Reduction

Clearly, if A and B are equivalent under row and column permutations, then $R(A) = R(B)$ and $C(A) = C(B)$ since the row and column sets are preserved under these permutations. This proves that the set of all B such that $R(B) = R(A)$ and $C(B) = C(A)$ contains all matrices equivalent to A under row and column permutations (i.e. the set of all B such that $R(B) =$

²It may be helpful to brush up on the theory of vector spaces, but you may choose to not bother with the finer details and take my word for it!

³Refer Appendix D.4

⁴This question was taken from Codeforces. The editorial indicates that my hashing solution wasn't intended, but when has that ever stopped me?

$R(A)$ and $C(B) = C(A)$ is a superset of the set of all B equivalent to A under row and column permutations).

Now notice that the number of row and column permutations on A are $n! \cdot m!$ (including the identity permutation). Each of these permutations gives a different matrix since the entries are all distinct. Also it can be shown that the number of matrices with given $R(M)$ and $C(M)$ is also $n! \cdot m!$ (In fact while proving this you will notice that the fact we are proving is quite intuitive)

Since the two sets have the same size and one contains the other, they must be equal. Hence $R(A) = R(B)$ and $C(A) = C(B)$ implies that A and B are equivalent under row and column permutations.

(Note that we assumed all entries are distinct in the above proof)

This reduced problem can be solved by two approaches:

4.4.1 Method 1: Set of Sets Approach (Row and Column sets)

Formally, let

$$R_i = \{a_{i1}, a_{i2}, \dots, a_{im}\}$$

be the representation of row i . We compare:

$$\{R_1, R_2, \dots, R_n\}_A = \{R_1, R_2, \dots, R_n\}_B$$

and similarly for columns.

This method is conceptually exact but computationally heavy due to the need to compare complex set-of-sets structures. Set structures can be implemented using balanced binary trees in $\Theta(n \log n)$ time - In C++ we have the `set` data structure which is implemented as a balanced binary tree. Comparing two sets of size m takes $\Theta(m \log m)$ time, and we have to do this for n rows and n columns, leading to a total time complexity of $\Theta(nm \log n \log m)$.

C++ implementation:

```
//Matrices are 0-indexed and stored in 2D vectors "first" and "second"
set<set<int>>>F;//set of sets for first matrix
for(int i=0;i<n;i++)
{
    set<int>temp(first[i].begin(),first[i].end()); //set of ith row
    F.insert(temp);
}

for(int i=0;i<n;i++)
{
    set<int>temp(second[i].begin(),second[i].end()); //set of ith row
    if (!F.count(temp)) //early exit if this row doesn't exist in the first matrix
    {
        cout<<"NO\n";
        return;
    }
}
```

4.4.2 Method 2: Hashing Based Verification of Rows and Columns

To obtain a more efficient solution, each row is hashed to a pair of values (sum, squaresum), where sum and squaresum denote the sum and sum of squares of elements in the row respectively. This representation is much easier to compare across rows.

Assuming the hash is sufficiently collision-resistant, we compare the sets of row hashes for both matrices. This can be implemented efficiently using `unordered_set` in C++, which provides expected $\Theta(1)$ insertion and lookup time.

For each row i , we compute:

$$h(R_i) = \left(\sum_{j=1}^m a_{ij}, \sum_{j=1}^m a_{ij}^2 \right)$$

We define a custom hash for a pair (x, y) as:

$$H(x, y) = x \cdot 2^{32} + y$$

where x and y are the computed row statistics. This choice of pair hash is popular for 32 bit integers, as it combines the two values into a single 64-bit integer guaranteeing unique hashes.

We insert the hashes of all rows of matrix A into an unordered set and then check if the hashes of rows of matrix B are present in this set. If any hash from B is not found in the set from A , we can conclude that the matrices are not equivalent under row permutations.

Similarly for columns, we compute the column hashes and compare them in the same manner.

This method has an expected time complexity of $\Theta(nm)$ due to the linear time required to compute the hashes and the expected constant time for hash set operations.

This method involved hashing so the correctness of the algorithm is itself a random variable – but given the constraint that the matrix entries are from $\{1, 2, \dots, nm\}$, it seems highly unlikely that this will fail. In fact, it seems that this may be a unique hash in this case, but i am yet to find a proof for general (n, m) .

C++ implementation:

```
auto pair_hash=[](int x,int y){
    return ((x)<<32)+y;
};

unordered_set<long long>s;
loop(i,n)
{
    long long v1=0,v2=0;
    loop(j,m)
    {
        int x=first[i][j];
        v1+=x;
        v2+=x*x;
    }
    s.insert(pair_hash(v1,v2));
}
loop(i,n)
{
    long long v1=0,v2=0;
    loop(j,m)
    {
        int x=second[i][j];
        v1+=x;
        v2+=x*x;
    }
```

```

    }
    if(!s.count(pair_hash(v1,v2))) //early exit if this row doesn't exist in the first matrix
    {
        cout<<"NO\n";
        return;
    }
}

```

4.5 A Sliding Window Problem

The Problem Two integer arrays x and y , each of size n , are given along with an integer k ($1 \leq k \leq n$). For every contiguous subarray (window) of size k , determine whether the corresponding windows in x and y contain the same multiset of elements at every position.

More precisely, for every index (1-based) i such that $1 \leq i \leq n - k + 1$, we compare the multisets:

$$\{x_i, x_{i+1}, \dots, x_{i+k-1}\} \quad \text{and} \quad \{y_i, y_{i+1}, \dots, y_{i+k-1}\}$$

and check whether they are identical for all i .

Rolling Hash Method

To obtain an efficient solution, we maintain two rolling hashes for each window:

$$S = \sum a_i, \quad Q = \sum a_i^2$$

These values can be updated in $\Theta(1)$ time when the window slides by removing the outgoing element and adding the incoming element.

For each position i , we compute:

$$(S_x(i), Q_x(i)) \quad \text{and} \quad (S_y(i), Q_y(i))$$

and check whether:

$$(S_x(i), Q_x(i)) = (S_y(i), Q_y(i))$$

If the condition holds for all i , we conclude that every corresponding window has identical multiset structure.

C++ Implementation

```

long long sumX=0,sumY=0;
long long sumX2=0,sumY2=0;

for(int i=0;i<k;i++)
{
    sumX+=x[i];
    sumY+=y[i];
    sumX2+=x[i]*x[i];
    sumY2+=y[i]*y[i];
}

bool ok=1;
for(int i=k;i<n;i++)
{

```

```

if(!((sumX==sumY)&&(sumX2==sumY2)))
{
    ok=0;
    break;
}

sumX-=x[i-k];
sumX+=x[i];
sumX2-=x[i-k]*x[i-k];
sumX2+=x[i]*x[i];

sumY-=y[i-k];
sumY+=y[i];
sumY2-=y[i-k]*y[i-k];
sumY2+=y[i]*y[i];
}
cout<<(ok?"YES":"NO");

```

This method is very easy to implement and runs in $\Theta(n)$ time, which is efficient for this problem. What was especially nice about this method is that we can maintain the hash values in constant time as the window slides, which is what I meant by using a **rolling hash**. However, it relies on the assumption that the hash values are unique for different multisets. We can improve this (if needed) by doing a second round of hashing with a different hash function like polynomial hashing.

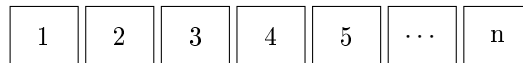
Chapter 5

Data Structures

5.1 Skip Lists

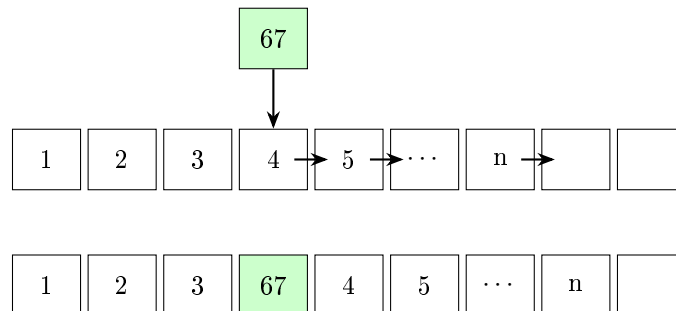
Why not just use Arrays?

Suppose we store the following array in contiguous memory:



Suppose we wish to insert the element 67 between 3 and 4.

Since arrays occupy contiguous memory, every element beginning from 4 must be shifted one position to the right in order to create space for the new element.

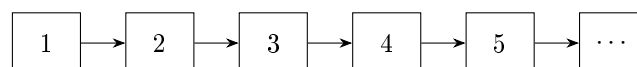


Insertion into an array requires shifting $\Theta(n)$ elements in the worst as well as average case, so a single insertion is $\Theta(n)$ time complexity.

How are Linked Lists any better?

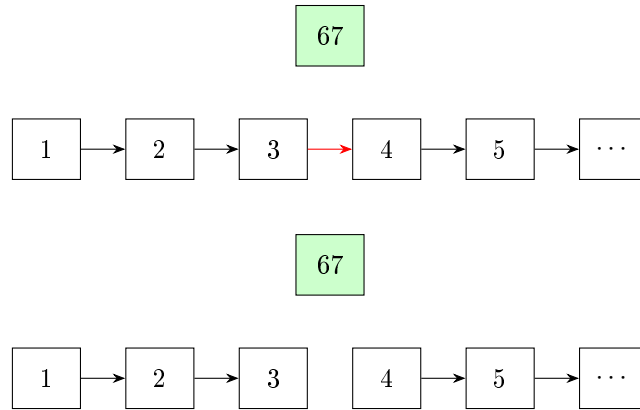
Unlike arrays, linked lists do not store elements contiguously in memory. Instead, each element stores $\{\text{Value}, \text{Next}\}$, where **Next** is a pointer to the next element.

So a linked list containing the elements $1, 2, 3, 4, 5, \dots, n$ may be represented as

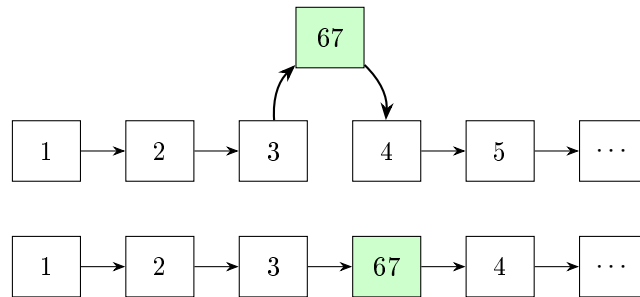


To insert an element 67 between say 3 and 4, we simply create a new node containing 67 and link it between 3 and 4.

Step 1: Remove the old connection



Step 2: Create new connections



Done!

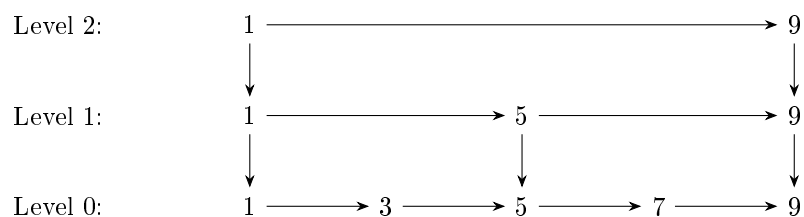
We used a constant number of pointer operations to insert an element into the list. Thus insertion is $\Theta(1)$ time complexity.

Why Skip Lists?

Although insertion is $\Theta(1)$ time complexity, searching for an element in linked lists is $\Theta(n)$ time complexity because we have to traverse the whole list. Even if the linked list is "sorted", we can not binary search since that requires random access in constant time, which a linked list can not provide.

So the idea of skip lists is to bypass that limitation by adding “express lanes” to a linked list. Instead of traversing every element one-by-one, we maintain additional higher-level lists that skip¹ over many elements at once.

Consider the following skip list –



Higher levels contain fewer elements and allow us to jump across large portions of the list quickly.

¹Hence the name!

5.1.1 Expected Structure of a Skip List

The logarithmic complexity of skip lists arises from the probabilistic structure of the levels. Recall that:

- every node appears in level 0,
- each node independently gets promoted to the next level with probability $\frac{1}{2}$.

Consequently:

$$\Pr(\text{node reaches level } k) = \frac{1}{2^k}$$

Thus, the expected number of nodes in level k is:

$$\frac{n}{2^k}$$

Level	Expected Number of Nodes
0	n
1	$n/2$
2	$n/4$
3	$n/8$
$\vdots$	$\vdots$

5.1.2 Expected Height

Let H be the height of a skip list with n independent nodes, where each node is promoted to the next level with probability $1/2$. Then:

$$\Pr(H \geq h) = 1 - (1 - 2^{-h})^n.$$

We use the tail-sum formula:

$$\mathbb{E}[H] = \sum_{h=1}^{\infty} \Pr(H \geq h) = \sum_{h=1}^{\infty} \left(1 - (1 - 2^{-h})^n\right).$$

Step 1: Split the expectation Let $h_o = \lfloor \log_2 n \rfloor$

$$\mathbb{E}[H] = \sum_{h \leq h_o} \Pr(H \geq h) + \sum_{h > h_o} \Pr(H \geq h) =: S_1 + S_2.$$

Step 2: Bounding S_1

$$\text{Let } \Pr(H \geq h) = f(h) = 1 - (1 - 2^{-h})^n \implies f(h_o) \leq f(h) \leq f(0) \forall 0 \leq h \leq h_o$$

$$\implies \sum_1^{h_o} f(h_o) \leq \sum_1^{h_o} f(h) \leq \sum_1^{h_o} f(0)$$

Since $f(0) = 1$, and $f(h_o) \approx 1 - (1 - \frac{1}{n})^n \approx 1 - \frac{1}{e}$ we get:

$$h_o(1 - \frac{1}{e}) \leq S_1 \leq h_o$$

Step 3: Bounding S_2 Since $(1 - x)^n \geq 1 - nx$,

$$1 - (1 - 2^{-h})^n \leq n \cdot 2^{-h}.$$

$$\implies \sum_{h > h_0} n \cdot 2^{-h} \geq S_2.$$

Since LHS of that inequality evaluates to ≈ 2

$$\implies S_2 \leq 3$$

Step 4: Combining bounds

$$(1 - e^{-1})h_0 + 3 \leq \mathbb{E}[H] \leq h_0 + 3$$

Since $h_0 \approx \log_2 n$, we get:

$$\boxed{\mathbb{E}[H] = \Theta(\log n)}$$

Geometric Decline Theorem

Recall that using the markov inequality, we proved that the height of the recursion tree for **QuickSelect** is atmost $\Theta(\log n)$. Using the same technique, we can show that if a sequence of nonnegative random variables satisfies

$$\frac{\mathbb{E}[X_{i+1}]}{\mathbb{E}[X_i]} \leq p \quad (0 < p < 1),$$

and T denotes the first index for which $X_T \leq 1$, then

$$\mathbb{E}[T] \leq \Theta(\log(X_0)),$$

Applying the Theorem

Let X_i denote the number of nodes present at level i of the skip list. Since every node is promoted independently with probability $1/2$,

$$\mathbb{E}[X_i] = n \left(\frac{1}{2}\right)^i. \implies \frac{\mathbb{E}[X_{i+1}]}{\mathbb{E}[X_i]} = \frac{1}{2}.$$

The height of the skip list is precisely the highest level containing at least one node. Equivalently, if

$$H = \min\{i : X_i \leq 1\},$$

then the height is H .

Applying the Geometric-Decline theorem with $p = \frac{1}{2}$ gives

$$\mathbb{E}[H] \leq \Theta(\log n).$$

Why did we do the first method at all?

In the second method we did not prove a lower bound! Although even in further analyses we won't need the lower bound, it is good to have established that it is exactly $\Theta(\log n)$. Also, we revised the "splitting" technique for finding the complexity of an expression, which is a useful tool in general.

5.1.3 Searching in a Skip List

Suppose we wish to search for the element 7.

We begin at the top left level and move right while the next element is ≤ 7 . Once that element is strictly greater than 7, we move down to the next level. Repeat until we reach the bottom level.

Pseudocode:

SEARCH(x):

current = top-left node

while current exists:

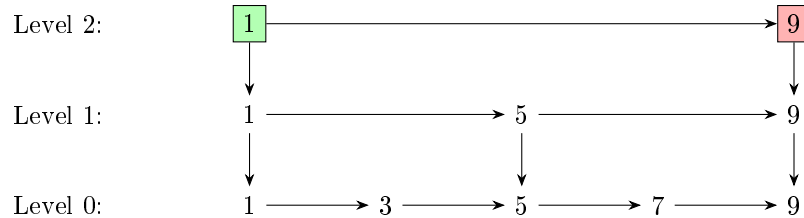
 while current.right exists
 and current.right.value $\leq$ x:
 current = current.right

 if current.value == x:
 return FOUND

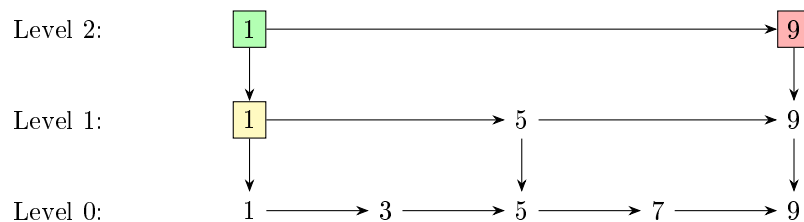
 current = current.down

return NOT FOUND

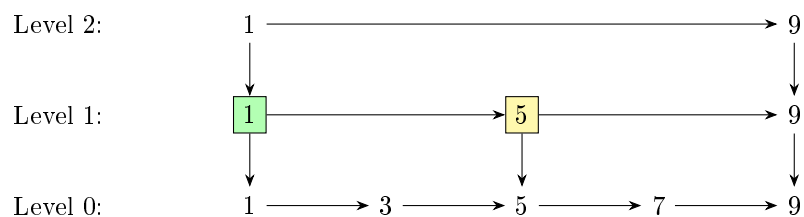
Step 1: Start at top, check right



Since on the right we have an element strictly greater than 7, we intend to move down.

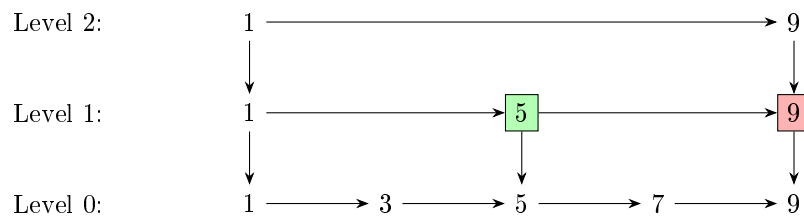


Step 2: Moved down, now check right

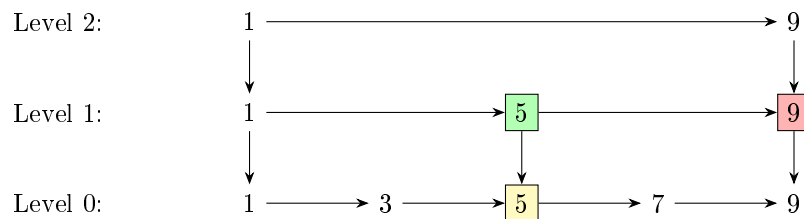


Since $5 \leq 7$, we intend to move right.

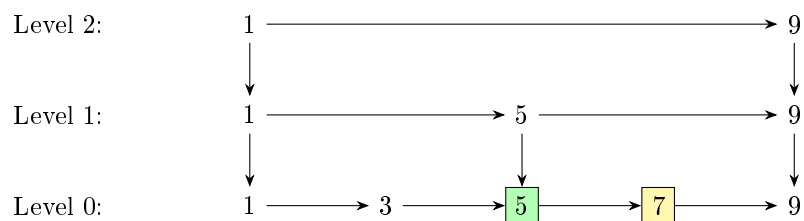
Step 3: Moved right, check right



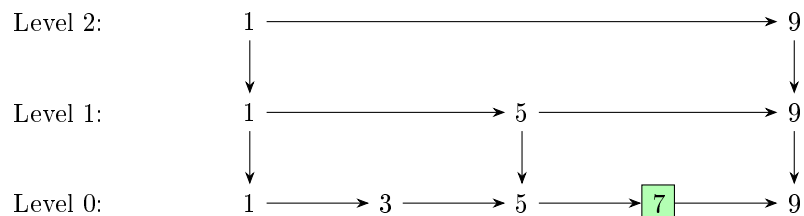
Since $9 > 7$, we intend to move down.



Step 4: Moved down, check right



Step 5: Moved right, found the target!



We move right to 7 and locate the desired element.

5.1.4 Randomised, But Why?

One may wonder why skip lists use randomization at all. Could we not deterministically place:

- every second element in level 1,
- every fourth element in level 2,
- every eighth element in level 3,
- and so on?

While this would indeed support logarithmic search complexity, insertions and deletions would become expensive.

For example, inserting a new element near the beginning of the list changes which elements are the second, fourth, eighth, ... elements of the list.

Consequently, many levels may require rebuilding.

Randomization avoids this problem by assigning heights independently to each node. Thus, updates are $\Theta(\text{height}) = \Theta(\log n)$ while the structure stays balanced in expectation.

Thus, randomized skip lists achieve logarithmic expected search *and* insert complexity without requiring complicated balancing operations such as rotations²!

5.1.5 Insertions

The insertion operation in a skip list closely mirrors the search procedure. In fact, insertion is essentially a search followed by a series of pointer modifications.

To insert a key x , we first perform the standard search starting from the topmost level. At each level, we move right until we either find x or encounter the first node with a key greater than x (or reach the end of the list).

If the key x is found during this traversal, then we do nothing. Otherwise, the search naturally terminates at the position where x should logically appear in the bottom level. This position is determined by the first “blocking” node whose key is greater than x , or by reaching a null pointer at the end of the list.

Once this position is identified, insertion proceeds as follows. We insert x immediately to the left of the blocking node at level 0. After placing it in the bottom layer, we decide randomly (typically with probability $1/2$) whether the node should also appear in higher levels. If it is promoted, we move upward level by level, and at each level we again insert it in the same relative horizontal position, maintaining alignment with the structure discovered during the search phase.

Thus, the update process reuses the search path and only modifies pointers locally along that path. This is what ensures that skip list insertion remains logarithmic in expected time.

5.2 The Hash Table/Unordered Set

The need arose for a data structure to store a set of elements with efficient search, insertion and deletion operations – so we turn to the hash table!

Initialisation – Empty Array We begin with an empty array of size m . Since this is a plain old normal array of size m , we can access and modify any random element in constant time ...

0	1	2	3	4	5	6	7	
								...

We add an element x by computing $i = \mathbf{H}(x)$. For example, suppose $\mathbf{H}(6) = 3$. So we place 6 at index 3.

First insertion

0	1	2	3	4	5	6	7	
			6					...

Similarly, we continue inserting elements in the same way as long as no two elements collide at the same index.

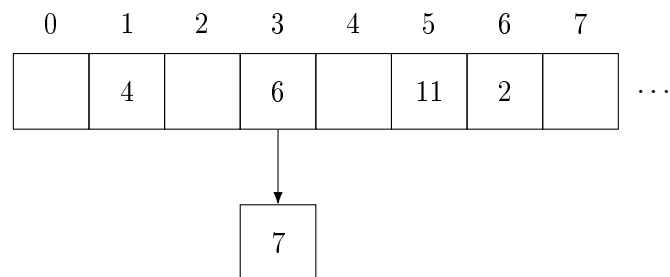
²Structures such as balanced binary trees require balancing operations e.g. in the implementation of `set` in C++ STL library

More insertions

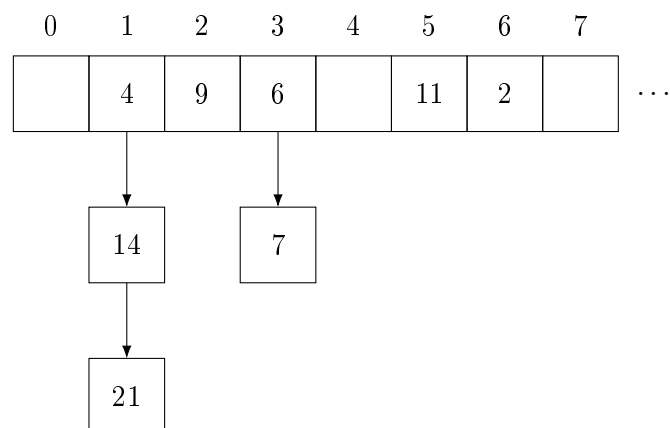
0	1	2	3	4	5	6	7	
	4		6		11	2		...

At this point, every index contains at most one element.

First *collision* Now we attempt to insert 7. Suppose $\mathbf{H}(7) = 3$. We try to place 7 at index 3, but we find 6 already present. So we extend that position into a linked list.



More collisions As more elements are inserted, collisions may occur at multiple indices, forming multiple linked lists.



Thus, each index i stores a linked list of all elements inserted till now who hash to i .

Supported Operations

```
Insert(x)
{
    i = H(x)
    traverse Linked_List[i]
    {
        if x present
            return
    }
    add x to end of Linked_List[i]
}
Search(x)
{
    i = H(x)
    traverse Linked_List[i]
    {
        if x found
            return true
    }
    return false
}
Delete(x)
{
    i = H(x)
    traverse Linked_List[i]
    if x found
        remove x from Linked_List[i]
}
```

Complexity Analysis

Let:

- n be the number of elements stored
- m be the number of indices in the array

For a uniform hash function, the next element could go uniformly randomly at any of the m indices. The expected length of the linked list (and thus the time required to do any operation) is:

$$\frac{\sum_{i=1}^m L_i}{m} = \frac{n}{m}.$$

Therefore:

- Search takes expected time $O\left(\frac{n}{m}\right)$
- Insertion takes expected time $O\left(\frac{n}{m}\right)$
- Deletion takes expected time $O\left(\frac{n}{m}\right)$

If $m = \Theta(n)$, then:

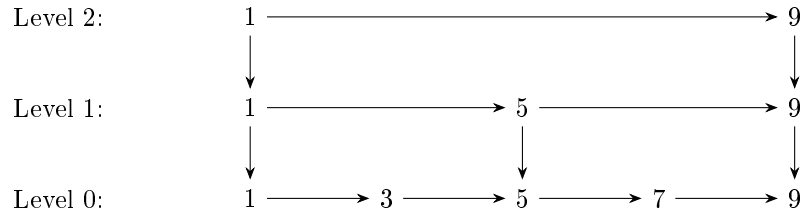
$$\frac{n}{m} = \Theta(1),$$

so all operations take expected $\Theta(1)$ time.

We maintain the condition that $m = \Theta(n)$ by resizing and rehashing if n is too large.

5.3 Problems

- 1 (a) Prove that the expected search time in a skip list is $\Theta(\log n)$. *Hint: downward moves are bounded by height, which is expected $\Theta(\log n)$.*
- (b) Deduce similarly that insertion takes expected $\Theta(\log n)$ time.
- (c) Try inserting 8 into the following skip list using the procedure we described earlier:



- (d) Can you think of a $\Theta(\log n)$ deletion strategy for skip lists?
- 2 (a) For hashes of form $H(x) = (ax + b) \bmod c$, why is it preferred to have c as a prime number? *Hint: Think of the arithmetic progression (a very common occurrence in data) with common difference d – what happens when d is not coprime with c ?*

Chapter 6

Number-Theoretic Algorithms

6.1 Introduction

Prime numbers have fascinated mathematicians for thousands of years. They appear deceptively simple to define – a prime is a positive integer greater than one whose only divisors are 1 and itself. Yet many of the deepest questions in mathematics revolve around the distribution and properties of primes.

Beyond their theoretical significance, prime numbers occupy a central position in modern computing. Public-key cryptographic systems such as **RSA** rely on arithmetic involving very large primes, often hundreds or thousands of bits long. As a result, computers routinely face questions such as

- Is a given integer prime?
- How can a large prime be generated efficiently?
- If an integer is composite, can its factors be found quickly?

At first glance, these problems appear daunting. Consider an integer with several hundred decimal digits. Testing divisibility by every smaller integer is clearly infeasible, and even more sophisticated deterministic approaches may require substantial computational effort.

Surprisingly, randomness provides an elegant alternative. Instead of attempting to prove directly that a number is prime, randomized algorithms often search for evidence that it is composite. A carefully chosen random experiment can reveal such evidence with high probability.

6.2 The Fermat Test

The simplest algorithm to test the primality of an integer n is to check for divisibility by every integer from 2 up to $\sqrt{n}$. Assuming division can be performed in constant time, this algorithm runs in $\Theta(\sqrt{n})$ time. Any algorithm we come up with should be at least as efficient as this!

The central result underlying our first algorithm is Fermat's Little Theorem. Let the set of primes be denoted by $\mathbb{P}$ and the set $\{0, 1, \dots\}$ by $\mathbb{N}$.

6.2.1 Fermat's Little Theorem

$$p \in \mathbb{P}, a \in \mathbb{N} \implies a^p \equiv a \pmod{p}$$

A useful equivalent form is obtained if $p \nmid a$, in which case we can divide both sides by a to get

$$a^{p-1} \equiv 1 \pmod{p}$$

This observation immediately suggests a test for primality. Given an integer n , choose some value a , compute $a^{n-1} \bmod n$, and check whether the congruence holds. If the congruence fails, then n cannot be prime.

6.2.2 The Algorithm

```
exp(a, x, n) // computes a^x modulo n in O(log x)
{
    if x == 0
        return 1
    y = exp(a, x/2, n)
    if x even
        return (y*y) mod n
    else
        return (a*y*y) mod n
}

PrimalityTest(n) // for n>=4
{
    if(n even)
        return "composite"

    choose a random a in {2, ..., n-2}

    /*We avoid a=1 and a=n-1 since they trivially satisfy
    the congruence even when n is composite */

    if exp(a, n-1, n) != 1
        return "composite"
    else
        return "probably prime"
}
```

Note the rather beautiful technique used in the function `exp(a, x, n)`¹. Another way of looking at it is that we find the binary expansion of x and compute the powers of a corresponding to the bits of x using repeated squaring.

6.2.3 Error Probability

The natural question is now obvious –

"How often can a composite number fool the test?"

For many composite integers the probability is quite small. Unfortunately, there exists a remarkable family of numbers that pass Fermat's test for *every* admissible choice of a .

¹This is sometimes called the binary exponentiation algorithm.

These numbers are known as Carmichael numbers², and they expose a fundamental weakness of the algorithm.

Fermat's test merely checks whether the final result of a^{p-1} equals 1. It ignores the sequence of intermediate values that arise during repeated squaring.

6.3 The Miller–Rabin Test

6.3.1 A Key Property

Let p be an odd prime and let d such that

$$p - 1 = 2^s d, \quad d \text{ is odd}$$

Define the sequence $x_i = a^{2^i d}, i \in \mathbb{N}$. So

$$x_0 = a^d, \quad x_1 = a^{2d}, \quad x_2 = a^{4d}, \quad \dots, \quad x_s = a^{2^s d}.$$

By Fermat's Little Theorem, we have $x_s = a^{p-1} \equiv 1 \pmod{p}$.

Clearly, each term is the square of the previous one modulo p . That is,

$$x_{i+1} \equiv x_i^2 \pmod{p}$$

A simple fact about prime moduli –

$$p \in \mathbb{P}, x \in \mathbb{N}, x^2 \equiv 1 \pmod{p} \implies x \equiv 1 \pmod{p} \text{ or } x \equiv -1 \pmod{p}.$$

So either $x_0 \equiv 1 \pmod{p}$ or there is some index $i \geq 0$ such that $x_i \equiv -1 \pmod{p}$.

²The smallest Carmichael number is 561. It has been proven that there are infinitely many Carmichael numbers.

6.3.2 The Algorithm

```
exp(a, x, n)
{
    if x == 0
        return 1
    y = exp(a, x/2, n)
    if x even
        return (y*y) mod n
    else
        return (a*y*y) mod n
}

isPrime(n) // for n>=4
{
    if(n even)
        return "composite"

    s=0
    d=n-1
    while d is even
        increment s by 1, divide d by 2

    //here we have decomposed n-1 into 2^s and d, where d is odd

    Choose a random a in {2, ..., n-2}
    x = exp(a, d, n)

    if x == 1 or x == n-1
        return "probably prime"

    loop i=1 to s-1
    {
        x = (x*x) mod n
        if x == n-1
            return "probably prime"
        if x == 1
        {
            return "composite"
            /*since hitting 1 mod n early means we have found a
            non-trivial square root of 1 mod n, so n is composite*/
        }
    }
    return "composite"
}
```

6.3.3 Error Probability

It can be shown that if n is composite, then the probability that the Miller–Rabin test returns "probably prime" is at most $1/4$.

The proof³ that the Miller–Rabin test has error probability at most $1/4$ is quite technical and relies on some non-trivial number theory and group theory. Let $T = \{2, \dots, n-2\}$, the set of all possible a 's. The key and final step is to show that one can partition T into disjoint subsets of size at least 4 such that each subset contains at most one false positive. Thus,

$$|F| \leq |T|/4 \implies \frac{|F|}{|T|} \leq \frac{1}{4}$$

6.3.4 Performance

For large n , we can no longer say that arithmetic operations take constant time, and the time complexity of the multiplications involved are d^2 where d is the number of digits in n , which is $\Theta(\log n)$. Since there are $\Theta(\log n)$ multiplications involved in binary exponentiation, total time complexity is $\Theta(\log^3 n)$.

Repeating the test k times will take $\Theta(k \log^3 n)$ time. A reasonable choice of k (e.g., $k = 34$) makes the error probability at most 10^{-20} .

6.4 Pollard's Rho Algorithm

The Problem – Given a composite integer n , find a non-trivial⁴ factor of n .

The Solution – We can create a pseudo-random sequence of integers $x_0, x_1, \dots, x_i, \dots$ such that $x_i \equiv x_{i-1}^2 + c \pmod{n}$ for some c .

If at any point $\gcd(x_i - x_j, n)$ is not 1 or n , we can take that to be a factor of n by definition⁵ of the gcd!

Why do we use the weird sequence $x_i \equiv x_{i-1}^2 + c \pmod{n}$? Why not generate a sequence of random integers and do the same thing? The reason behind using a deterministic sequence is that in this iterative process, we can use Floyd's cycle-finding algorithm to detect collision which takes time linear in the cycle size. In the other case, we would have to store and check with all previous values of x_j for every new x_i generated. That is quadratic in the length of the cycle! Secondly, Choosing x_0 and c randomly acts as a seed for the pseudo-random process. The sequence is thus overall truly random for our analysis purposes yet compatible with Floyd's algorithm.

³In case you are interested, see <https://codeforces.com/blog/entry/98102>

⁴In this case, that would be a factor other than 1 and n

⁵Conventionally, $\gcd(0, n)$ is taken to be n .

6.4.1 Floyd's Cycle-Detection Algorithm

```
FloydCycleDetection(f,x)
{
    tortoise = f(x)
    hare = f(f(x))

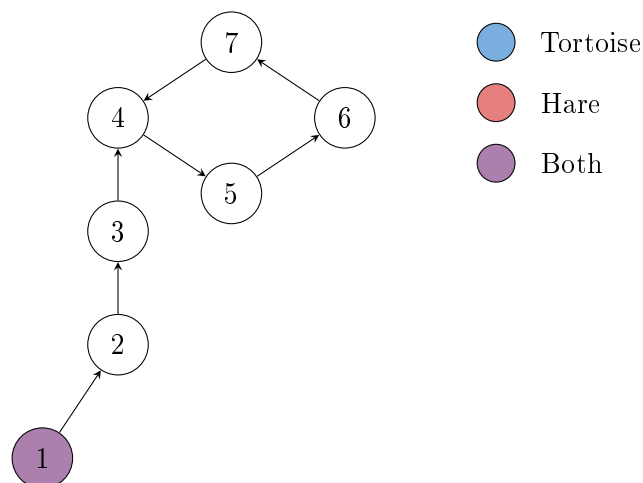
    while tortoise != hare:
    {
        tortoise = f(tortoise)
        hare = f(f(hare))
    }

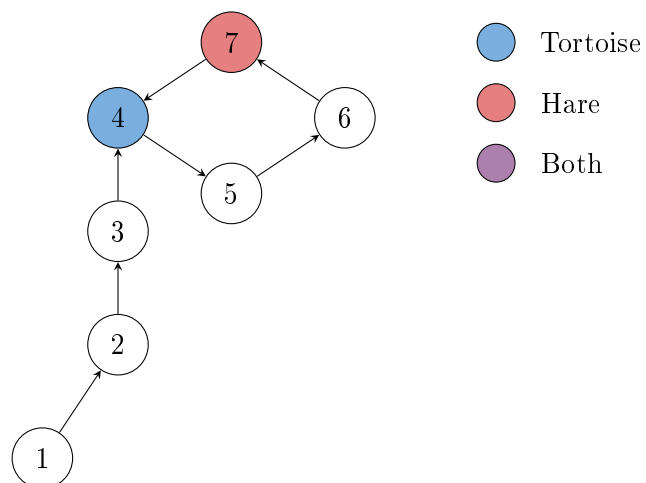
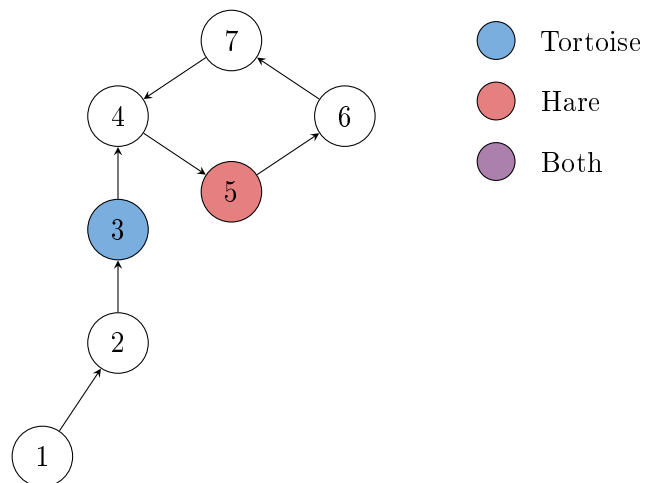
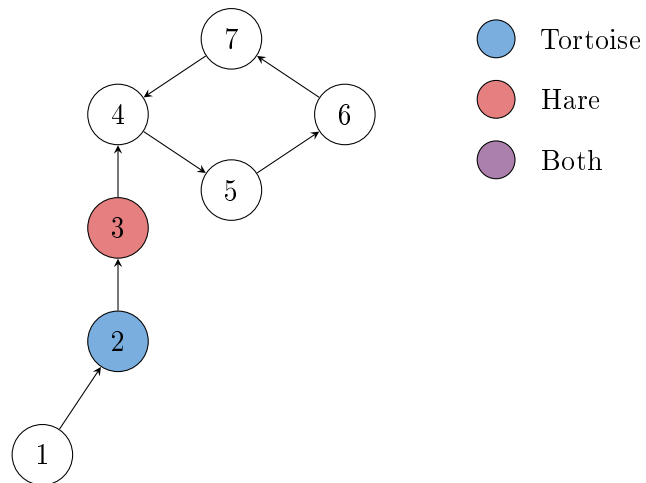
    return
}
```

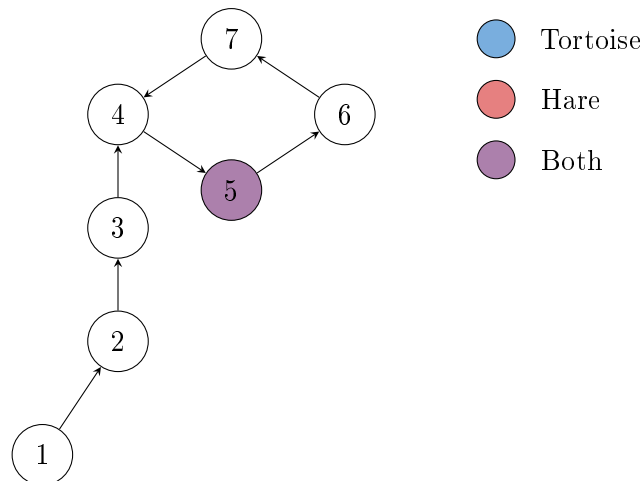
Example

We can represent the sequence with a directed graph as follows – there is an edge between each x and $f(x)$ or equivalently each x_i and x_{i+1} . The starting node is x_0 where both the hare and tortoise start. It can be proven that Floyd's algorithm works with some effort, but the point of this example is to give a visual intuition rather than write a rigorous proof. Also, I like colorful graphs!

The shape of the graph is deliberately chosen to be like the greek letter ρ , which is where the name **Pollard's Rho** comes from. It is the general shape of the sequence which the algorithm considers, because the output space of $f(x)$ is finite.







And thus when both coincide, we can report we have found the cycle! Try this on a cycle with odd number of nodes to really convince yourself that it works.

6.4.2 The Algorithm

```

f(t)
{
    return (t*t + c) mod n
}

PollardRho(n)
{
    if n is even:
        return 2

    choose random c in [1, n-1]
    choose random x in [0, n-1]

    y = x
    d = 1

    while d == 1:
    {
        x = f(x)
        y = f(f(y))

        d = gcd(|x - y|, n)
    }

    if d == n:
        restart algorithm with new x, c

    return d
}

```


6.4.3 Expected Running Time Analysis

Collision Model

Our sequence $x_0, x_1, x_2, \dots$ is generated by iterating a polynomial function modulo n . Since the choice of x_0 and c is random, the sequence can be treated as independent samples from a uniform distribution over $\{0, 1, \dots, n-1\}$, until repetition occurs. And we are looking for a hidden number p , the modulo over which the sequence repeats. Since p divides n , the sequence when taken modulo p will thus act like independent samples from a uniform distribution over $\{0, 1, \dots, p-1\}$.

Equivalently it can be said that the distribution $\text{Uniform}(0, n-1)$ modulo p is equivalent to the distribution $\text{Uniform}(0, p-1)$.

This naturally leads us to the birthday paradox⁶ model, where we are interested in the expected time until a collision occurs in a sequence of random samples from a finite set.

Suppose we define the collision time T as the smallest index such that a repetition occurs,

$$\iff T = \min\{j \mid x_i = x_j \text{ for some } i < j\}$$

Probability of No Collision

Let P_k denote the probability that the first k samples are all distinct. This is equivalent to $\Pr(T > k)$. Then using the fact that the samples are uniform and independent, we can express P_k as

$$P_k = \Pr(T > k) = \prod_{i=0}^{k-1} \left(1 - \frac{i}{p}\right)$$

Expected Collision Time

The expected collision time can be expressed as

$$\mathbb{E}[T] = \sum_{k=0}^{\infty} \Pr(T > k) = \sum_{k=0}^{\infty} P_k = 1 + \left(1 - \frac{1}{p}\right) + \sum_{k=2}^{\infty} P_k = 2 - \frac{1}{p} + \sum_{k=2}^{\infty} P_k$$

We will use $2 - \frac{1}{p} \leq 2$ and the inequality $1 - x \leq e^{-x}$ to bound this.

$$\begin{aligned} P_k &= \prod_{i=0}^{k-1} \left(1 - \frac{i}{p}\right) \leq \prod_{i=0}^{k-1} e^{-i/p} = e^{-\sum_{i=0}^{k-1} i/p} = e^{-k(k-1)/(2p)} \\ \implies \mathbb{E}[T] &\leq 2 + \sum_{k=2}^{\infty} e^{-k(k-1)/(2p)} \end{aligned}$$

For $k \geq 2$,

$$\begin{aligned} k(k-1) &\geq \frac{k^2}{2} \implies \frac{k(k-1)}{2p} \geq \frac{k^2}{4p} \implies e^{-k(k-1)/(2p)} \leq e^{-k^2/(4p)} \\ \implies \mathbb{E}[T] &\leq 2 + \sum_{k=2}^{\infty} e^{-k^2/(4p)} \end{aligned}$$

⁶The probability that in a group of 23 people we find two with the same birthday is approximately 1/2. This is of course, not a true *paradox* but simply a surprising result from an intuition standpoint.

Since the function $f(x) = e^{-x^2/(4p)}$ is monotonically decreasing on $[0, \infty)$, we can compare the sum with an integral!

$$\sum_{k=2}^{\infty} e^{-k^2/(4p)} \leq \int_0^{\infty} e^{-x^2/(4p)} dx.$$

Using the substitution

$$\begin{aligned} x &= 2\sqrt{p}t, & dx &= 2\sqrt{p}dt, \\ \implies \int_0^{\infty} e^{-x^2/(4p)} dx &= 2\sqrt{p} \int_0^{\infty} e^{-t^2} dt = 2\sqrt{p} \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\pi p}. \\ &\implies \mathbb{E}[T] \leq 2 + \sqrt{\pi p} \end{aligned}$$

So the expected collision time is at most $\Theta(\sqrt{p})$.

Collision Detection?

Once a collision occurs modulo p , say

$$x_i \equiv x_j \pmod{p}, \quad i \neq j,$$

we know p must divide $x_i - x_j$. But how do we check that if we don't know p ? Knowing we want to find a p which divides $x_i - x_j$ and n , the logical step is to find the $\gcd(x_i - x_j, n)$, because if that is 1 then p is definitely⁷ 1. And as soon as $\gcd(x_i - x_j, n)$ becomes not 1 (and not n), we can take that as the divisor. We know all p for which the collision happened are definitely⁸ smaller than this number so we could not have done better!

Due to Floyd's cycle detection algorithm, we don't need to keep track of all pairs of indices i, j . We can simply keep track of the pairs (x_i, x_{2i}) for $i = 1, 2, \dots$ and check for collisions between these pairs. This is what the algorithm does by maintaining two pointers x and y that move at different speeds.

Cost per Iteration

Each step of the algorithm performs a constant number of modular multiplications and one greatest common divisor computation. Using the Euclidean algorithm,

$$\gcd(a, b)$$

can be computed in at most $\Theta(\log \max(a, b))$ time.

Hence each iteration costs at most $\Theta(\log n)$ time since n is the largest number we are working with for all the involved operations.

Total Expected Complexity

Since a collision modulo the smallest prime factor p occurs after $\Theta(\sqrt{p})$ steps, the expected running time is

$$\Theta(\sqrt{p} \log n).$$

If p is the smallest prime divisor of n , this dominates the runtime. In the worst case where $p \approx n^{1/2}$, we obtain the heuristic bound $\Theta(n^{1/4} \log n)$.

⁷This is concluded using the lemma all common divisors of a and b divide the $\gcd(a, b)$

⁸By definition of the $\gcd$, it must be the *greatest* common divisor!

Chapter 7

The Probabilistic Method

7.1 The Dominating Set Problem

The Problem Given a graph:

$$G = (V, E)$$

find a smallest possible subset:

$$D \subseteq V$$

such that every vertex in the graph is either:

- contained in D , or
- adjacent to some vertex in D .

Such a set is called a *dominating set*.

Intuitively, we may think of the vertices in D as “control centers” or “facilities” such that every location in the graph is either itself a facility or directly connected to one.

Unfortunately, finding the minimum dominating set is NP-hard, so we want to find a polynomial time algorithm that gives us a reasonably small¹ dominating set.

A Randomised Solution

Suppose the graph has minimum degree d . We now consider the following extremely simple randomised strategy:

Choose k vertices randomly out of the n vertices (uniformly among the $\binom{n}{k}$ possible choices). If the chosen set forms a dominating set, output it. Otherwise, try again!

At first glance, this algorithm seems dumb. However, it turns out that if we choose k large enough, the probability of failure becomes small. The key question thus becomes:

What is the probability of failure as a function of k , n , and d ?

We now analyse this probability.

¹We care more about the asymptotic behaviour of the size of set produced by the algorithm and less about which polynomial time complexity the algorithm has.

Probability That a Vertex Remains Undominated

Fix a vertex $v \in V$. Since the graph has minimum degree d , $N[v]$ contains at least $d+1$ vertices. For v to remain undominated, none of the selected k vertices may lie inside $N[v]$

When choosing vertices uniformly at random, the probability that a single chosen vertex does *not* belong to $N[v]$ is at most:

$$1 - \frac{d+1}{n}$$

$$\implies \Pr[v \text{ remains undominated}] \leq \left(1 - \frac{d+1}{n}\right)^k$$

Using the inequality² $1 - x \leq e^{-x}$ for all $x \in \mathbb{R}$, we obtain:

$$\Pr[v \text{ remains undominated}] \leq e^{-k(d+1)/n}$$

Bounding the Failure Probability

The algorithm fails if at least one vertex remains undominated.

Applying the union bound³

$$\Pr[\text{failure}] \leq n \cdot e^{-k(d+1)/n}$$

We would like this probability to be strictly less than 1. Therefore we require:

$$ne^{-k(d+1)/n} < 1$$

$$\implies \ln n - \frac{k(d+1)}{n} < 0$$

$$\implies k > \frac{n \ln n}{d+1}$$

Thus, choosing:

$$k = \frac{cn \ln n}{d+1}$$

for any constant $c > 1$ gives $\Pr[\text{failure}] \leq n^{1-c}$ which tends to 0 as $n \rightarrow \infty$.

What Does This Actually Mean?

The important point is not merely that the algorithm succeeds with high probability.

Rather, we have shown something stronger –

There exists a dominating set⁴ of size roughly:

$$\frac{n \log n}{d}$$

Indeed, if no such dominating set existed, our algorithm could never possibly succeed.

Thus a simple probabilistic argument simultaneously:

- proves the existence of a reasonably small dominating set,
- and gives a randomised algorithm capable of finding one.

²Refer Appendix B.1

³Refer Appendix A.2

⁴This method has a limitation - if $d < c \ln n$, then the size of the dominating set produced by the algorithm is basically the whole set of vertices, making it useless.

Amplifying the Success Probability

Even if a single run fails, we may simply repeat the algorithm independently.

If the failure probability of one run is $p = \frac{1}{n^{c-1}}$, then after t independent repetitions, the probability that *every* run fails equals p^t which decreases exponentially fast.

So the tradeoff is choosing a larger c has a smaller expected number of repetitions $(\frac{n^{c-1}}{n^{c-1}-1})^5$ but a larger dominating set, while choosing a smaller c has a larger expected number of repetitions but a smaller dominating set.

7.2 Bipartite Subgraph

Problem statement. Given an undirected graph $G = (V, E)$ with $|E| = m$, we wish to find a bipartition of V that maximizes the number of edges between vertices of different partitions. Equivalently, we want a 2-coloring of the vertices that maximizes the number of bichromatic edges. We aim to show that we can always guarantee at least $m/2$ edges in the cut.

7.2.1 Expected Number of Bichromatic Edges

Let X be the number of bichromatic edges under a random 2-coloring of the vertices. Each edge has a probability $\frac{1}{2}$ of being bichromatic, so define the indicator variable I_i as 1 when the edge numbered i is bichromatic and 0 otherwise. We can conclude that for a random vertex 2-coloring, $\Pr(I_i = 1) = \frac{1}{2}$ and $\Pr(I_i = 0) = \frac{1}{2}$, so $\mathbb{E}[I_i] = \frac{1}{2}$. Since $X = \sum_{i=1}^m I_i$, we have:

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^m I_i\right] = \sum_{i=1}^m \mathbb{E}[I_i] = \frac{m}{2}$$

A Weak Lower Bound

Let $X \in \{0, 1, \dots, m\}$ be the number of bichromatic edges under a random 2-coloring of the vertices. We already have $\mathbb{E}[X] = \frac{m}{2}$.

We derive a crude lower bound on $\Pr(X \geq \frac{m}{2})$.

Case 1: $m = 2k$ (even).

Let $p = \Pr(X \geq k)$. The maximum value of $E(X)$ under these constraints would be achieved when we push all mass in $\{0, 1, \dots, k-1\}$ to $k-1$, and all mass in $\{k, \dots, 2k\}$ to $2k$. Then

$$\mathbb{E}[X] \leq (1-p)(k-1) + p(2k).$$

Using $\mathbb{E}[X] = k$,

$$k \leq (1-p)(k-1) + 2kp$$

$$k \leq k-1 + p(k+1)$$

$$1 \leq p(k+1)$$

$$p \geq \frac{1}{k+1} = \frac{2}{m+2}.$$

$$\implies \Pr\left(X \geq \frac{m}{2}\right) \geq \frac{2}{m+2}.$$

⁵Since the probability of failure is $p = (n^{c-1} - 1)/n^{c-1}$ and this is a bernoulli trial, the expected number of trials for success is $1/p$

Case 2: $m = 2k + 1$ (odd).

Let $p = \Pr(X \geq k + 1)$. The maximum value of $E(X)$ under these constraints would be achieved when all mass in $\{0, 1, \dots, k\}$ is pushed to k , and all mass in $\{k + 1, \dots, 2k + 1\}$ is pushed to $2k + 1$. Then

$$\mathbb{E}[X] \leq (1 - p)k + p(2k + 1).$$

Using $\mathbb{E}[X] = k + \frac{1}{2}$,

$$k + \frac{1}{2} \leq k + p(k + 1)$$

$$\frac{1}{2} \leq p(k + 1)$$

$$p \geq \frac{1}{2(k + 1)} = \frac{1}{m + 1}.$$

$$\implies \Pr\left(X \geq \frac{m}{2}\right) \geq \frac{1}{m + 1}.$$

Conclusion –

$$\Pr\left(X \geq \frac{m}{2}\right) \geq \min\left(\frac{2}{m + 2}, \frac{1}{m + 1}\right) = \frac{1}{m + 1}.$$

7.2.2 A Simple Randomised Algorithm

From the probabilistic analysis, a random coloring satisfies

$$\Pr\left(X \geq \frac{m}{2}\right) \geq \frac{1}{m + 1}.$$

Therefore, if we repeatedly sample random colorings independently, the probability that none of T trials succeeds is at most

$$\left(1 - \frac{1}{m + 1}\right)^T.$$

Choosing $T = (m + 1) \ln(m + 1)$ gives

$$\left(1 - \frac{1}{m + 1}\right)^T \leq e^{-\frac{T}{m + 1}} = e^{-\ln(m + 1)} = \frac{1}{m + 1}$$

Which goes to 0 as $m \rightarrow \infty$.

While simple, this method does not exploit structure of intermediate solutions and relies purely on repeated random sampling rather than improvement steps.

7.2.3 Derandomisation

Instead of making random choices, we assign vertices one by one smartly, trying to maximise the number of bichromatic edges. Suppose some vertices have already been assigned and the remaining vertices are still unassigned. Let X be the random variable denoting the number of bichromatic edges. Let L and R be the bipartitions.

Denote the expected value of X given the first i assignments by:

$$\mathbb{E}[X \mid i]$$

For the $i + 1$ -th assignment of vertex v , we have two choices:

- place v in L ,

- place v in R .

The current conditional expectation is the average of the expectations obtained from these two choices. Therefore, at least one choice must yield a conditional expectation that is at least as large as the current one.

$$\max(\mathbb{E}[X \mid i, v \text{ in } L], \mathbb{E}[X \mid i, v \text{ in } R]) \geq \frac{1}{2}(\mathbb{E}[X \mid i, v \text{ in } L] + \mathbb{E}[X \mid i, v \text{ in } R]) = \mathbb{E}[X \mid i]$$

We simply assign the next vertex such that it would add the most bichromatic edges in this step – that choice yields a better expectation for the future.

$$\text{Since I have ensured that } \mathbb{E}[X \mid i + 1] \geq \mathbb{E}[X \mid i]$$

$$\implies \mathbb{E}[X \mid n \text{ assignments}] \geq \mathbb{E}[X \mid 0 \text{ assignments}]$$

$$\implies X_{final} \geq \frac{m}{2}$$

Why did that work?

Initially $\mathbb{E}[X] = \frac{m}{2}$. At every step we choose an assignment that does not decrease the conditional expectation⁶. Hence the conditional expectation never drops below $m/2$!

After all vertices have been assigned, no randomness remains. The conditional expectation is then equal⁷ to the actual number of crossing edges produced by the algorithm.

$$\implies X_{final} \geq \frac{m}{2}.$$

Thus we obtain a deterministic algorithm that always constructs a bipartite subgraph containing at least half of the edges.

Why does this not work everytime?

The properties of this particular problem we needed for this to work were –

1. $\mathbb{E}[X \mid 1] = m/2$, regardless of which assignment that was – symmetry between the left and right sides.
2. It was obvious which assignment would maximise $\mathbb{E}[X \mid i + 1]$ given $\mathbb{E}[X \mid i]$.

Both these properties do not hold, for example, in the Dominating Set problem. Depending on your first choice (to put in the dominating set or dominated set), you might get a very good dominating set in expectation or a very bad one. An extreme counter example is a star shaped⁸ graph, where if you put the central node in dominated set you will get a dominating set of size $n - 1$, but if you put it in dominating set you will get a dominating set of size 1. The second property also fails because in further choices it is not clear how to choose which set to put the node in, since we have no idea how to determine any expectation of this problem.

⁶This is by no means a proof for optimality – it does not maximise the number of bichromatic edges.

⁷If you are feeling tricked somehow, I relate fully! But it is true, we designed this deterministic algorithm and simultaneously proved that it produces at least $m/2$ crossing edges using conditional expectation!

⁸A star shaped graph is a graph with one central node and $n - 1$ nodes connected to it and only it.

Chapter 8

Online Decision Algorithms

What is an Online Algorithm?

An online algorithm is an algorithm that processes its input piece-by-piece in a serial fashion, without having the entire input available from the start. In some problems, this doesn't matter – the obvious solution is online since it requires a single pass over the array only. But in some problems, the online requirement makes the problem much harder and forces us to make decisions without knowing the future.

Online decision algorithms have randomness involved in their analysis for non-adversarial¹ inputs. Further, the goal of the algorithm usually is not deterministic optimality, but rather to maximise the probability of success or to achieve a good competitive ratio².

8.1 The Dating Problem

The Problem Consider the problem of finding the best partner. Assume that the number of people you will eventually date is fixed, i.e. some known value n . The people appear in a uniformly random order. After dating a person, you must immediately decide whether to commit to them forever or reject them permanently and continue searching. Figure out a strategy that maximises the probability of ending up with the best partner.³

We assume that while dating person i , we only know how they compare relative to the people we have already dated. So we could sort the partners till now in a total order if needed.

Solution: Most course materials, internet reels or shorts I have encountered this problem in directly jump to the strategy of:

“Reject the first r people, then accept the first person afterwards who is better than everyone seen so far.”

However, they often omit the proof of why an optimal strategy must necessarily have this threshold form. We first prove this fact.

Optimality Proof

Suppose we are currently considering the k -th person.

¹They are often called oblivious adversaries

²Competitive ratio is a measure of how well an online algorithm performs compared to an optimal offline algorithm that has access to the entire input from the start. The term n -competitive is used when an algorithm achieves at least $\frac{1}{n}$ times the "performance" of an optimal offline algorithm.

³This is a different problem from maximising the expected “value” of your partner. Here the objective is specifically to maximise the probability of selecting the single best partner among all n people.

If the current person is *not* the best among the first k people seen so far, then accepting them can never succeed: This is because there already exists an earlier person who is strictly better, and that earlier person has already been rejected. Hence an optimal strategy should only ever consider accepting people who are the best seen so far.

Now suppose the current person *is* the best among the first k people seen so far. Let q_k be the optimal probability of eventually selecting the globally best person if we reject the k -th person and continue optimally afterwards.

Q] If we instead accept the current person immediately, what is the probability of success?

A] Since the ordering of the n people is uniformly random, each of the first k positions is equally likely to contain the globally best person. Conditioned on the fact that the current person is the best among the first k , the probability that they are actually the globally best person equals $\frac{k}{n}$.

Therefore, accepting gives success probability $\frac{k}{n}$, rejecting gives success probability q_k . The optimal decision is to accept iff $\frac{k}{n} \geq q_k$ ⁴.

Now observe that $\frac{k}{n}$ is **increasing** with k .

On the other hand, q_k is **decreasing** with k . To see this, notice that from time k , one valid strategy is:

“Reject one more person no matter what, and then behave optimally from time $k+1$ onwards.”

This strategy achieves success probability exactly q_{k+1} . Since q_k is the *optimal* achievable success probability starting from time k , we must have:

$$q_k \geq q_{k+1}$$

Since $\frac{k}{n}$ is increasing and q_k is decreasing, the optimal decision changes at most once from rejecting to accepting as k increases. Therefore there exists some threshold $0 \leq r \leq n$ such that:

- for $k \leq r$, rejecting is optimal,
- for $k > r$, accepting the first “best-so-far” person is optimal.

Hence every optimal strategy must be a threshold strategy. Note that $r = 0$ corresponds to accepting the first person, and $r = n$ corresponds to rejecting everyone and ending up with no partner. Both of these are valid threshold strategies, though they are not very good ones⁵.

Finding the Optimal Threshold

We now analyse the threshold strategy:

“Reject the first r people, then accept the first person afterwards who is better than everyone seen so far.”

⁴This is a common theme in online algorithms - the optimal decision at each step is determined by comparing the expected “value” of actions and choosing the one with maximum value. Here our “value” is the success probability. In game theory this is called “utility” instead.

⁵Though it may be argued that having no partner is the best choice - this is beyond the scope of this text and is left as an exercise for the reader.

Suppose the globally best person appears at position k . For our strategy to succeed:

1. We must have $k > r$
2. Among the first $k - 1$ people, the best person must appear within the first r positions.

The first condition ensures that we do not automatically reject the best person. The second condition ensures that no earlier candidate after position r triggers acceptance before the globally best person appears.

$$\Pr[\text{best person appears at position } k] = \frac{1}{n}$$

Further, among the first $k - 1$ people, each position is equally likely to contain their maximum.

$$\implies \Pr[\text{maximum among first } k - 1 \text{ lies in first } r] = \frac{r}{k - 1}$$

$$\implies \Pr[\text{success}] = \sum_{k=r+1}^n \frac{1}{n} \cdot \frac{r}{k - 1} = \frac{r}{n} \sum_{k=r}^{n-1} \frac{1}{k} = \frac{r}{n} (H(n - 1) - H(r - 1))$$

Using the approximation $H(x) \approx \ln(x)$,

$$\frac{r}{n} (H(n - 1) - H(r - 1)) \approx \frac{r}{n} \ln \left(\frac{n - 1}{r - 1} \right) \approx \frac{r}{n} \ln \left(\frac{n}{r} \right)$$

$$x = \frac{r}{n} \implies P(x) = x \ln \left(\frac{1}{x} \right)$$

$$\implies P'(x) = \ln \left(\frac{1}{x} \right) - 1$$

$$P'(x_o) = 0 \implies \ln \left(\frac{1}{x_o} \right) = 1 \implies x_o = \frac{1}{e}$$

Since $\frac{r_o}{n} = \frac{1}{e} \implies r_o = \frac{n}{e}$, the optimal strategy is –

“Reject approximately the first $\frac{n}{e}$ people, then accept the first person afterwards who is better than everyone seen so far.”

The corresponding success probability is $\frac{1}{e} \ln e = \frac{1}{e}$ which is approximately **36.8%**. Not bad at all!

Strictly speaking, we should also verify that this critical point indeed corresponds to a maximum rather than a minimum. One way is via the second derivative test –

$$P''(x) = -\frac{1}{x} < 0, \forall x \in \mathbb{R}^{>0}$$

so $P(x)$ is concave and hence the critical point is necessarily a global maximum.

Alternatively, we can avoid calculus entirely by inspecting the behaviour near the boundaries.

$$\because P(x) = x \ln \left(\frac{1}{x} \right)$$

As $x \rightarrow 0$, $P(x) \rightarrow 0$ since x decays faster than $\ln(1/x)$ grows.

Similarly, as $x \rightarrow 1$, $P(x) \rightarrow 0$ because $\ln(1) = 0$.

Since $P(x)$ is positive for $0 < x < 1$, rises away from 0, and eventually falls back to 0, the unique critical point at

$$x = \frac{1}{e}$$

must correspond to the global maximum.

Behaviour for Small n

We used some limiting n approximations in finding the optimal threshold (I hope you noticed!). In case you were not planning on dating a tending-to-infinite number of people, you might be interested in the optimal strategy for small values of n . For finite n , the exact success probability for threshold r is:

$$\max_{0 \leq r \leq n} \left(\frac{r}{n} \sum_{k=r}^{n-1} \frac{1}{k} \right)$$

Computationally we can now easily find max for each n :

n	2	3	4	5	6	7
n/e	0.74	1.10	1.47	1.84	2.21	2.58
$\text{round}(n/e)$	1	1	1	2	2	3
Actual optimal r	1	1	1	2	2	3

The corresponding optimal success probabilities are:

n	2	3	4	5	6	7
$\text{Pr}[\text{success}]$	0.500	0.500	0.458	0.433	0.428	0.414

We observe that the optimal threshold follows the value suggested by $\frac{n}{e}$ even for relatively small values of n , and it will get closer and closer as n increases. The success probability is already around 41.4% for $n = 7$, and it will approach the limiting value of $\frac{1}{e} \approx 36.8\%$ as n grows.

8.2 The Parallel Processing Problem

Consider a setting with m identical machines processing incoming tasks. The tasks arrive one after another, and the arrival times are sufficiently close that we may assume they arrive within negligible time of each other.

Each incoming task must immediately be assigned to one of the m machines. Once a decision is made, it cannot be reversed. Thus, the algorithm has no knowledge of future tasks while making its decision.

The tasks assigned to a machine are processed sequentially and hence form a queue at that machine. Suppose the incoming tasks have processing times $T_1, T_2, \dots, T_n$ where n itself is not known in advance.

The objective is to minimize the *makespan*, i.e. the finishing time of the busiest machine:

$$C_{\max} = \max_{1 \leq i \leq m} T_i$$

where T_i denotes the total processing time assigned to machine i . This is a natural objective since the overall processing time is determined by the machine that finishes last⁶.

8.2.1 Greedy Algorithm

A natural strategy is the following greedy algorithm.

```

Assign(Task T)
{
    find machine with min load
    assign T to this machine
}

```

Performance Analysis

Interestingly, we do not actually have a polynomial time algorithm to find the optimal *offline* solution to this problem.

We will therefore use lower bounds on OPT (the optimal offline makespan) to derive an upper bound on the competitive ratio of the greedy algorithm.

Let ALG denote the makespan produced by the greedy algorithm. Consider the task that finishes last in the greedy schedule. This may or may not be the last *assigned* task. Let its processing time be p .

Immediately before assigning this job, suppose the minimum machine load was L . By our hypothesis that this machine will be the one that finishes last, the greedy makespan is

$$ALG = L + p$$

Hence the total load already assigned before this final job was at least mL . Including the final job, the total processing load is therefore at least $mL + p$.

By Pigeonhole Principle, at least one machine must have load at least the average load, so –

$$OPT \geq \frac{mL + p}{m} = L + \frac{p}{m}$$

Since any valid schedule must process the job of size p ,

$$OPT \geq p \implies OPT\left(1 - \frac{1}{m}\right) \geq p\left(1 - \frac{1}{m}\right)$$

Adding the two inequalities gives:

$$OPT\left(2 - \frac{1}{m}\right) \geq L + p = ALG \implies \boxed{\frac{ALG}{OPT} \leq 2 - \frac{1}{m}}$$

Therefore the greedy algorithm is $(2 - \frac{1}{m})$ -competitive, sometimes written as 2-competitive since the difference is negligible for large m .

⁶This concept is similar to the Rate Determining Step (RDS) in the field of Chemical Kinetics

8.2.2 Tightness of the Bound

The previous analysis showed that the greedy algorithm is $(2 - \frac{1}{m})$ -competitive. We now construct an example showing that the analysis cannot, in general, be improved beyond a factor of $(2 - \frac{1}{m})$.

Consider $m = 4$ machines and the following sequence of incoming jobs:

$$1, 1, 1, \dots, 1 \text{ (12 times)}, 4$$

The greedy algorithm will assign the first 12 jobs as follows:

Machine i	1	2	3	4
Jobs assigned	1, 1, 1	1, 1, 1	1, 1, 1	1, 1, 1

After processing these 12 jobs, each machine has load 3. The last job of size 4 will be assigned to any machine (say machine 1), resulting in a final makespan of 7. On the other hand, an optimal offline algorithm could have assigned the first 12 jobs as follows:

Machine i	1	2	3	4
Jobs assigned	1, 1, 1, 1	1, 1, 1, 1	1, 1, 1, 1	4

After processing these 13 jobs, each machine has load 4. Thus the optimal makespan is 4, and the competitive ratio is $\frac{7}{4}$, which is indeed our bound of $2 - \frac{1}{4}$.

Chapter 9

Heuristic Algorithms

Many computational problems of practical interest have a statement like “Given a graph find a set of vertices of size at most x satisfying property P .”

Now one way to guarantee a solution is to find the minimum sized set of vertices satisfying P , but this is usually not doable in polynomial time.

So we are often forced to settle for a solution that is not necessarily optimal but can be found efficiently. Such algorithms are *heuristic algorithms* – algorithms designed to quickly find solutions that are often good in practice, even if they do not always provide provable optimality guarantees or even any guarantees at all.

Inherently the analysis of heuristic algorithms involve probability and expectation, making them a natural application of the tools we have developed so far.

Notice that the greedy algorithm we saw in the previous chapter was a heuristic algorithm as well!

9.1 The Dominating Set Problem Revisited

Recall the dominating set problem: given a graph $G = (V, E)$, find a smallest possible subset $D \subseteq V$ such that every vertex in the graph is either contained in D or adjacent to some vertex in D .

A Greedy Heuristic

A very natural approach is the following greedy algorithm:

Repeatedly select the vertex that dominates the largest number of currently undominated vertices.

More precisely:

1. Initially, all vertices are marked undominated.
2. At each step, choose a vertex whose closed neighbourhood $N[v]$ contains the largest number of undominated vertices.
3. Add this vertex to the dominating set.
4. Mark all vertices in $N[v]$ as dominated.
5. Repeat until every vertex becomes dominated.

At every step we greedily maximise immediate progress.

Analysing the Greedy Heuristic

Let OPT denote the size of a minimum dominating set.

Suppose that at some stage of the algorithm there remain r undominated vertices.

Since the optimal solution uses only OPT vertices to dominate the entire graph, those same OPT vertices must also collectively dominate these remaining r vertices.

Therefore, by the pigeonhole principle, at least one of those optimal vertices must dominate at least $\frac{r}{\text{OPT}}$ of the currently undominated vertices.

But the greedy algorithm always chooses a vertex dominating as many undominated vertices as possible. Hence the greedy step must dominate at least $\frac{r}{\text{OPT}}$ new vertices.

Thus after one greedy step, the number of remaining undominated vertices becomes at most

$$r - \frac{r}{\text{OPT}} = r \left(1 - \frac{1}{\text{OPT}}\right)$$

If r_t denotes the number of undominated vertices after t greedy steps, then

$$r_{t+1} \leq r_t \left(1 - \frac{1}{\text{OPT}}\right)$$

Since initially $r_0 = n$, we obtain

$$r_t \leq n \left(1 - \frac{1}{\text{OPT}}\right)^t$$

We would like all vertices to become dominated, i.e. $r_t < 1$.

Since

$$r_t \leq n \left(1 - \frac{1}{\text{OPT}}\right)^t$$

it suffices to find the smallest t such that

$$n \left(1 - \frac{1}{\text{OPT}}\right)^t < 1$$

Rearranging:

$$\left(1 - \frac{1}{\text{OPT}}\right)^t < \frac{1}{n}$$

Taking logarithms:

$$t \ln \left(1 - \frac{1}{\text{OPT}}\right) < -\ln n$$

Since $\ln(1 - \frac{1}{\text{OPT}}) < 0$, dividing reverses the inequality:

$$t > \frac{\ln n}{-\ln \left(1 - \frac{1}{\text{OPT}}\right)}$$

Let S denote the size of the dominating set produced by the greedy algorithm. Then S is the smallest integer t satisfying the above inequality. Thus:

$$S = \left\lceil \frac{\ln n}{-\ln \left(1 - \frac{1}{\text{OPT}}\right)} \right\rceil \leq \frac{\ln n}{-\ln \left(1 - \frac{1}{\text{OPT}}\right)} + 1$$

Now using the inequality

$$-\ln(1 - x) \geq x$$

with $x = \frac{1}{\text{OPT}}$, we obtain

$$-\ln\left(1 - \frac{1}{\text{OPT}}\right) \geq \frac{1}{\text{OPT}}$$

Taking reciprocals reverses the inequality:

$$\begin{aligned} \frac{1}{-\ln\left(1 - \frac{1}{\text{OPT}}\right)} &\leq \text{OPT} \\ \implies S &\leq \ln n \cdot \frac{1}{-\ln\left(1 - \frac{1}{\text{OPT}}\right)} + 1 \\ \implies S &\leq \text{OPT} \ln n + 1 \end{aligned}$$

Hence the greedy algorithm produces a dominating set of size at most

$$\boxed{\text{OPT} \ln n + 1}$$

9.2 Exam Scheduling

The Problem We are given a set of exams and a set of students. Each student is enrolled in a subset of the exams. We wish to schedule as many exams as possible in the first time slot¹, subject to the constraint that no student should have two exams scheduled at the same time².

The goal is to find a largest possible subset of exams that can be scheduled simultaneously without any conflict.

Reduction

We model the problem using a graph $G = (V, E)$ where:

- each vertex represents an exam,
- an edge exists between two vertices if there is at least one student enrolled in both corresponding exams.

In this formulation, a feasible schedule corresponds to a set of vertices with no edges between them. Such a set is called an *independent set*.

Thus, the problem reduces to finding a maximum independent set in a graph.

9.2.1 Random Permutation

V is the set of all vertices. Maintain a set S - the independent set we are building. Pick a random vertex from V - delete it from V and if none of its neighbours are in S , put it in S . Keep doing this until V is empty. This guarantees forming a valid independent set.

Note that the order in which we pick vertices is a uniformly random permutation of the vertices.

¹this is different from the problem of the overall scheduling problem which is equivalent to finding the Chromatic Number, i.e. the largest number of colors needed to colour a graph which is equivalent to the minimum number of slots required to conduct the whole exam schedule. This is a simpler problem where we only care about the first time slot.

²Or you could just give them a Time Turner and hope for the best :)

Probability a Vertex is Selected

Fix a vertex v . Consider the set consisting of v and its neighbours $N(v)$. There are $d(v) + 1$ vertices in total.

Since all relative orderings are equally likely, the probability that v is the picked earliest among these vertices is:

$$\Pr[v \in S] = \frac{1}{d(v) + 1}$$

Expected Size of the Independent Set

Let I_v be the indicator random variable for the event $v \in S$. Then:

$$|S| = \sum_{v \in V} I_v$$

Taking expectation and using linearity:

$$\mathbb{E}[|S|] = \sum_{v \in V} \mathbb{E}[I_v] = \sum_{v \in V} \frac{1}{d(v) + 1}$$

Therefore, there exists an independent set of size at least:

$$\sum_{v \in V} \frac{1}{d(v) + 1}$$

Denoting the maximum independent set size by $\alpha(G)$, we have:

$$\alpha(G) \geq \sum_{v \in V} \frac{1}{d(v) + 1}$$

Using the Jensen's³ inequality on $1/(x+1)$, we have:

$$\sum_{v \in V} \frac{1}{d(v) + 1} \geq \frac{n}{d + 1}$$

where d is the average degree of the graph. Hence:

$$\alpha(G) \geq \frac{n}{d + 1}$$

9.2.2 Local Decision

We now describe an alternative randomized construction that produces a large independent set via local random choices.

Each vertex independently chooses a random value:

$$X_v = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

We construct a set S by including a vertex v if:

- $X_v = 1$, and
- for every neighbour $u \in N(v)$, $X_u = 0$.

³Refer Appendix B.2

Probability of Inclusion

Fix a vertex v . The probability that v is included in S is:

$$\Pr[v \in S] = p(1 - p)^{d(v)}$$

Therefore:

$$\mathbb{E}[|S|] = \sum_{v \in V} p(1 - p)^{d(v)}$$

Guarantee on Expected Size

We analyze the randomized construction where each vertex is selected independently with probability p , and a vertex v is included in the independent set S if it is selected and none of its neighbors are selected. Then:

$$\mathbb{E}[|S|] = \sum_{v \in V} p(1 - p)^{d(v)}.$$

Using convexity of $f(x) = (1 - p)^x$ and Jensen's inequality, we obtain:

$$\sum_{v \in V} (1 - p)^{d(v)} \geq n(1 - p)^d,$$

where $d = \frac{1}{n} \sum_{v \in V} d(v)$ is the average degree. Hence:

$$\mathbb{E}[|S|] \geq np(1 - p)^d.$$

We now optimize the function:

$$f(p) = p(1 - p)^d.$$

Differentiating:

$$f'(p) = (1 - p)^d - pd(1 - p)^{d-1} = (1 - p)^{d-1}((1 - p) - pd).$$

Setting $f'(p) = 0$, we obtain:

$$1 - p - pd = 0 \quad \Rightarrow \quad p = \frac{1}{d + 1}.$$

Substituting this optimal value gives:

$$\mathbb{E}[|S|] \geq n \cdot \frac{1}{d + 1} \left(1 - \frac{1}{d + 1}\right)^d = \frac{n}{d + 1} \left(\frac{d}{d + 1}\right)^d.$$

Using the standard bound $\left(1 - \frac{1}{d+1}\right)^d \geq e^{-d/(d+1)}$, we obtain:

$$\mathbb{E}[|S|] \geq \frac{n}{d + 1} \cdot e^{-d/(d+1)} \geq \frac{n}{e(d + 1)}.$$

This is a slightly weaker bound than the previous method, but it has the advantage of being local and easily parallelizable, as each vertex makes its decision independently based on local information.

Recall from the chapter on data structures that in Skip Lists we let each element locally decide how many levels to appear in. The reason for that was updation was quick since we did not have to change everybody's decision just because of the addition of a single element.

A similar reasoning can be applied here to justify the utility of a local decision algorithm.

9.3 Bipartite Subgraph Revisited

Recall the problem of finding a bipartition of the vertices that maximizes the number of bichromatic edges. We showed that a random coloring achieves at least $m/2$ bichromatic edges in expectation, and with some probability. Now we refine our approach by showing that we can always find a coloring with at least $m/2$ bichromatic edges via a self-correction procedure.

9.3.1 Self-Correction

Start with a random coloring of the vertices. If it has at least $m/2$ bichromatic edges, we are done.

Otherwise, let X be the number of bichromatic edges. We have $X < m/2$, so there are more monochromatic edges than bichromatic edges. Call a vertex v *bad* if it has more monochromatic incident edges than bichromatic incident edges. Otherwise it is *good*.

Claim: If the current cut has fewer than $m/2$ crossing edges, then a bad vertex must exist.

Proof idea: Each monochromatic edge contributes equally to two endpoints. So total number of monochromatic edges is half the sum of monochromatic degrees of vertices. If globally there are more monochromatic edges than crossing ones, by pigeonhole principle⁴, some vertex must be incident to more monochromatic than bichromatic edges.

Now consider flipping the color of a bad vertex v . Let $d_h(v)$ be the number of monochromatic (same-color) edges incident to v and $d_c(v)$ the number of bichromatic edges. Since $d_h(v) > d_c(v)$, flipping v increases X by

$$\Delta X = d_h(v) - d_c(v) > 0.$$

Thus, each flip strictly increases the number of bichromatic edges. This means that the number of steps is upper bounded by $m/2$.

Pseudocode:

```
initialize random coloring of vertices
compute X
```

```
while exists bad vertex v:
    flip color of v
    update X locally
```

Complexity analysis: Each flip increases X by at least 1, and $X \leq m$, so there are at most m flips.

If we maintain for each vertex the counts of monochromatic and bichromatic incident edges, each flip can be updated in $\Theta(\deg(v))$. To find a bad vertex we need atmost $\Theta(n)$ time. Over all flips, total time is atmost

$$O\left(\sum_{v \text{ flipped}} (\deg(v) + n)\right) = \Theta(mn + m).$$

Hence the full procedure runs in atmost ⁵ $\Theta(mn + m)$ after initialization.

⁴You may also prove this by contradiction – if all vertices are bad, then the total number of monochromatic edges is more than the total number of bichromatic edges by summing the inequality, contradicting the assumption.

⁵this can be improved to $\Theta(m \log n)$ by using priority queues with lazy updates – its a fun competitive programming exercise, I encourage you to try it!

Appendices

Appendix A

Probability and Expectation

A.1 Linearity of Expectation

For any random variables $X_1, X_2, \dots, X_n$ (not necessarily independent),

$$\mathbb{E} \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n \mathbb{E}[X_i].$$

This holds regardless of dependence between variables.

A.2 Union Bound

For any events $A_1, A_2, \dots, A_n$,

$$\Pr \left(\bigcup_{i=1}^n A_i \right) \leq \sum_{i=1}^n \Pr(A_i).$$

This is often used to upper bound the probability that at least one “bad event” occurs.

A.3 Independence of Expectation

If X and Y are independent random variables, then

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

A.4 Variance

For a random variable X , define the variance of X as

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

For any two random variables X and Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y) \quad (\text{where } \text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y])$$

If X and Y are independent, then $\text{Cov}(X, Y) = 0$, so

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

More generally, for any collection $X_1, \dots, X_n$,

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j).$$

A.5 Indicator Variables

For an event A , define the indicator random variable

$$I_A = \begin{cases} 1 & \text{if } A \text{ occurs,} \\ 0 & \text{otherwise.} \end{cases}$$

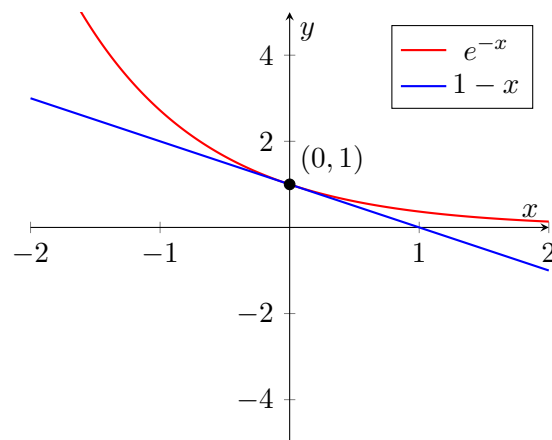
$$\implies \mathbb{E}[I_A] = \Pr(A)$$

Indicator variables are commonly used to convert counting problems into sums of random variables.

Appendix B

Inequalities

B.1 Exponential Inequality



$$1 - x \leq e^{-x} \quad \forall x \in \mathbb{R} \implies (1 - x)^k \leq e^{-kx} \quad \forall k \in \mathbb{N}, x \in \mathbb{R}$$

B.2 Jensen's Inequality

For a convex function f and random variable X ,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

Another form of Jensen's inequality is for a convex function f and real numbers $x_1, x_2, \dots, x_n$,

$$f\left(\frac{1}{n} \sum_{i=1}^n x_i\right) \leq \frac{1}{n} \sum_{i=1}^n f(x_i)$$

For concave functions, the inequality is reversed.

Appendix C

Discrete Mathematics

C.1 Pigeonhole Principle

Let $a_1, a_2, \dots, a_n$ be non-negative real numbers and let

$$S = \sum_{i=1}^n a_i.$$

Then there exists some index i such that

$$a_i \geq \frac{S}{n}.$$

Interpretation

This simply states that at least one value is at least the average value.

Proof by Contradiction

If every $a_i < \frac{S}{n}$, then summing over all i gives

$$\sum_{i=1}^n a_i < S,$$

which is a contradiction.

C.2 Definitions

A graph G is an ordered pair (V, E) where V is a set of vertices and $E \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is a set of edges.

Degree

For a vertex $v \in V$, the degree $d(v)$ is the number of edges which have v as an endpoint. Equivalently, $d(v)$ is the number of neighbours of v .

Given n is the number of vertices, the average degree of a graph is

$$d = \frac{1}{n} \sum_{v \in V} d(v)$$

Bipartite Graph

A graph $G = (V, E)$ is bipartite if the vertex set V can be partitioned into two disjoint sets L and R such that:

$$V = L \cup R, \quad L \cap R = \emptyset$$

and every edge connects a vertex in L to a vertex in R . That is, there are no edges with both endpoints in the same set.

Appendix D

Linear Algebra

D.1 Vector Spaces

Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set V equipped with vector addition and scalar multiplication satisfying the usual axioms:

- closure under addition and scalar multiplication,
- associativity and commutativity of addition,
- existence of a zero vector,
- existence of additive inverses,
- distributive properties,
- compatibility of scalar multiplication.

An example of a common vector space is the set of n -tuples of elements from $\mathbb{F}$,

$$\mathbb{F}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{F}\}$$

Common choices for the field $\mathbb{F}$ include the real numbers $\mathbb{R}$ or the integers $\mathbb{Z}$ with the usual addition and multiplication, and the finite field $\mathbb{F}_p$ where p is a prime with addition and multiplication modulo p .

D.2 Subspaces

A subset $W \subseteq V$ is called a subspace of the vector space V if

1. $0 \in W$
2. for all $u, v \in W$, we have $u + v \in W$
3. for all $u \in W$ and $\alpha \in \mathbb{F}$, we have $\alpha u \in W$

D.3 Dimension of a Subspace

A collection of vectors $v_1, v_2, \dots, v_k$ is called a basis of a subspace W if

- the vectors are linearly independent
- any vector in W is a linear combination of the vectors

The number of vectors in any basis of W is called the dimension of W and is denoted by $\dim(W)$.

D.4 The Nullspace

Let $M \in \mathbb{F}^{m \times n}$. The nullspace of M is defined by

$$\mathcal{N}(M) = \{x \in \mathbb{F}^n : Mx = 0\}.$$

The nullspace is also called the *kernel* of M , and we write

$$\ker(M) = \mathcal{N}(M).$$

The nullspace of a matrix is a subspace of $\mathbb{F}^n$.

D.5 Nullity and Rank

Since the nullspace is a subspace, it has a well-defined dimension.

Define $\text{nullity}(M) = \dim(\ker(M))$.

The rank of M , denoted $\text{rank}(M)$ is the dimension of the column space or row space of M , equivalently the maximum number of linearly independent columns or rows of M .

D.6 The Rank–Nullity Theorem

One of the fundamental results of linear algebra is the rank–nullity theorem.

Theorem (Rank–Nullity). For every matrix $M \in \mathbb{F}^{m \times n}$, we have

$$\text{rank}(M) + \text{nullity}(M) = n.$$

An immediate consequence is that if $M \neq 0$, then

$$\text{rank}(M) \geq 1, \implies \text{nullity}(M) \leq n - 1.$$